

Database Management Systems

UNIT-1

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Chapter 1: Introduction to Databases and Transactions



Relational Databases

A relational database (RDB) is a collective set of multiple data sets organized by tables, records and columns. RDBs establish a well-defined relationship between database tables. Tables communicate and share information, which facilitates data searchability, organization and reporting.

RDBs use Structured Query Language (SQL), which is a standard user application that provides an easy programming interface for database interaction.

Transaction Management

A transaction is one or more SQL statements that make up a unit of work performed against the database, and either all the statements in a transaction are committed as a unit or all the statements are rolled back as a unit. This unit of work typically satisfies a user request and ensures data integrity.

For example, when you use a computer to transfer money from one bank account to another, the request involves a transaction: updating values stored in the database for both accounts. For a transaction to be completed and database changes to be made permanent, a transaction must be completed in its entirety.

Chapter 2: Data Models

Chapter 2: Data Models



Data Model



Business Rules



Entity



Attribute



Relationships

One to One

One to Many

Many to Many

Business Rules



Database designer needs to take help from concepts such as entity, attributes and relationships to build a data model, but the above things are not sufficient to describe a system completely.



Business rules may define actors and prescribe how they should behave by setting constraints and help to manage business change in the system.



Business rules are statements of a discrete operational business policy or practice within specific organisations that constrains the business. It is intended to control or influence the behaviour of the business.



Examples of Business rules in Academia say University – Every student must have 75% attendance in order to sit for exam, A student has to pass 40 out of 100 in order to pass the subject, the maximum no of KT allowed per semester is 2 etc.

Chapter 2: Data Models

Hierarchical Model

- First DBMS model
- Data sorted hierarchically in top-down or bottom-up approach
- Uses pointers to navigate between stored data
- Represents data as a hierarchical tree
- One parent node can have many child nodes but one child cannot have more than one parent
- EXAMPLE – Information Management System (IMS) from IBM

Network Model

- Uses pointers but without the need of parent-child relationships.
- Used in network databases.
- Relationship between any two record types is called as SET.
- A relationship can be 1:N or M:N.
- EXAMPLE – IDS (Integrated Data Store) one of the products based on network models developed by IBM and North American Rockwell

Relational Model

- Proposed by E. F. Codd
- An attempt to simplify database structure by making use of tables and columns.
- Collection of 2-dimensional tables which consists of rows and columns.
- Tables are known as relations, columns are known as attributes and rows are known as tuples.
- Uses collection of tables to represent relationships amongst data.
- Each database item is viewed as a record with attributes. A record with similar attributes is called as table. Each table contains a record of particular type.
- The database uses relational model called as RDBMS (Relational Database Management System)

Entity Relationship Model

- Defines data elements and relationships among various data elements for a specified system.
- Based on the perception of real-world data that consists of set of entities (data items) and relationships among these entities.
- Popular high-level conceptual model used for conceptual design of database.
- An entity is a fundamental component of ER model which is a thing in real world with its own independent existence. Example Student, Department etc.
- Attributes are properties that describe the entities and its values become a major part of data stored in database.
- A particular entity will have some value for each of its attributes.
- Relationship is an association among several entities and is illustrated by diamond and read from left to right.

Object Model

- Data is stored in the form of objects which are structures called classes that display the data within it.
- Fields are instances of these classes called objects.
- Used in File Management System.
- DBMS developed is called OODBMS (Object Oriented Database Management System).
- OOD can handle complex applications such as scientific experiments, geographical information system etc.
- Draws its concept from real world objects which deals with data at higher level that is with the objects surrounding the data.
- Represents database in terms of objects, attributes and their behaviors.

Evolution of Data Models

GENERATION	TIME	MODEL	COMMENTS
First	1960s-1970s	File System	Used mainly on IBM mainframe system. Managed records, not relationships.
Second	1970s	Hierarchical and network	Early database systems. Navigational access.
Third	Mid-1970s to present	Object-oriented, Extended Relational	Support complex data. Extended relational products support objects and data warehousing.
Next Generation	Present to future	XML	Organization and management of unstructured data. Relational and object models add support for XML documents.

Chapter 3: Database Design and ER Diagrams

Goals of Database Design

01

To fulfill information requirements of the users and application.

02

To support all processing requirements as well as all performance criteria like response time, processing time and storage space.

03

Provide very simple and easy to understand structure of the information.

Overview of ER Model

In 1975 Chen developed the Entity Relationship (ER) model a high level data model that is useful in developing a conceptual design for a database.

Constructing ER diagram is a first step in designing a database.

ER model defines the data elements and relationships among them.

ER data model is based on perception of real world data consisting of entities (data items) and relationships among those entities.

Popular high level conceptual model used for conceptual design of database.

Entity

A fundamental component of ER model

It is a thing in real world with its own independent existence.
E.g. Student, faculty

May be an object with physical or logical existence

Has its own properties that describe the entity known as attributes



Attributes



Various properties that describe an entity



Attribute value that describes each entity becomes a major part of data stored in database as each entity will have some value for each of its attributes.



E.g. Employee entity has name, age, phone etc. as attributes.

Constraints On Relationships

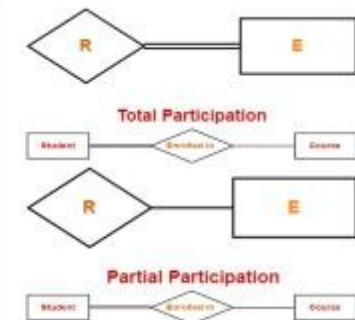
Mapping Constraints/ Cardinalities

ONE is to ONE – One tuple in entity is related only with one tuple in another entity. One department can have only one manager

ONE is to MANY – One tuple in entity is related with many tuples in another entity. One department can have many employees

MANY is to MANY – Many tuples in entity is related many tuples in another entity. Books in library issued by students

Constraints on Relationships



Chapter 3:EER Model

Extended Entity Relationship Model – ERD Issues



- EER model includes all modeling concept of ER model
- In addition it includes the concept of specialization and generalization
- EER model is used to represent collection of objects that is union of objects of different entity types
- A diagrammatic technique for displaying the EER concepts is called as EER diagrams

Specialization And Generalization

- Specialization
 - Top down approach of superclass/ subclass relationship
 - Process of defining a set of subclass of entity type called also as super class of specialization
 - Set of subclasses that forms a specialization is defined on the basis of some distinguishing characteristic of entity in super class
 - E.g. Set of subclass (Saving_Account, Current_Account) are specialization of superclass Account
 - The subclass defined in specialization is attached by lines to a circle which is connected to super class
 - Subset symbol on each line connecting a subclass to circle indicates the direction of super class/ subclass relationship
 - An attribute applied only to entities of particular subclass is called as specific attribute

Codd's Rule

- Information Rule – All available data in system should be represented as relations or tables.
- Guaranteed Access Rule – Each data item must be accessible without ambiguity by providing table name and its primary key of the row also include its column name to be accessed
- Systematic Treatment Of Null Values – Null values are not equal to blank space or zero they are unknown unassigned values which should be treated properly
- Self Describing Database – There should be dynamic online catalog based dictionary on relational model which keep information about tables data in database

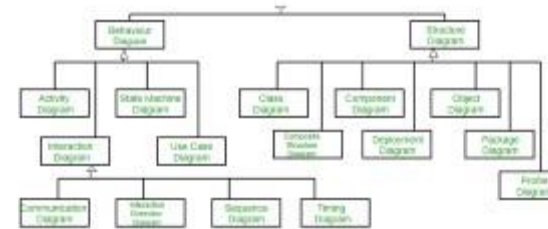
Chapter 3:UML Diagrams

UML(Unified Modelling Language)

- Introduction of UML
- Class Diagram
- Use-case Diagram
- Activity Diagram
- Implementation Diagram
- Advantages and Disadvantages

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Types Of UML Diagrams



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Class diagram

- Most popular UML diagrams used for construction of software applications.
- Similar to E-R diagram.
- It is a static diagram.
- Shown using class as its basic entity and lines between them represents relationship between them.
- Describes the attributes and operations of a class and also the constraints imposed on the system.

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Use case diagram

- Shows the interaction between users and the system in particular steps of tasks that users perform.
- Purpose is to capture the dynamic aspect of a system.
- Purpose is to gather requirements of a system and to identify external and internal factors influencing the system.

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Chapter 3:UML Diagrams

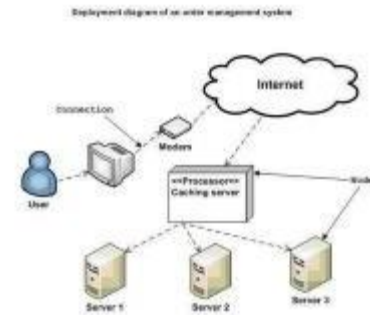
Activity Diagram

- Is basically a flowchart to represent the flow from one activity to another. The activity can be described as an operation of the system.
- It is a particular operation of the system.
- Are not only used for visualizing dynamic nature of a system but also used to construct the executable system by using forward and reverse engineering techniques.

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Implementation diagram

Are deployment diagrams which are used for describing about the hardware components where software components are deployed.



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Advantages Of UML Diagrams

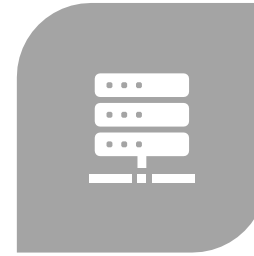
- It is the most useful method of visualization and documenting software systems design.
- It is effective for modeling large, complex software systems.
- It is simple to learn, but provides advanced features for expert analysts, engineers, designers and architects.
- It can specify systems in an implementation- independent manner.
- It specifies a skeleton that can be refined and extended with additional features.
- It specifies the functional requirements of system in an object oriented manner.

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Introduction to Databases and Transactions



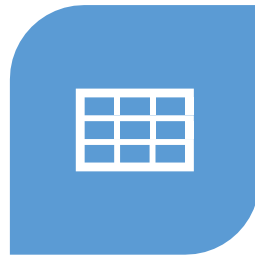
WHAT IS DATABASE
SYSTEM?



PURPOSE OF
DATABASE SYSTEM



VIEW OF DATA



RELATIONAL
DATABASES



DATABASE
ARCHITECTURE



TRANSACTION
MANAGEMENT

Data & Database & Database Management System

Data

The fact and figures that can be recorded in system and that have some special meaning assigned to it.

For Example: Data of a customer is name, telephone number, address, product purchased etc.

Database

A database is a collection of related data items stored at one place.

For example: College database stores information about students , teachers, classes, subjects(All related data).

DBMS is a software system that helps in the process of defining, constructing , manipulating the database.

It is also known as computerized record-keeping system.

Need of data in Database system



Data- As a corporate resource



Data Availability



Maintaining Complex data



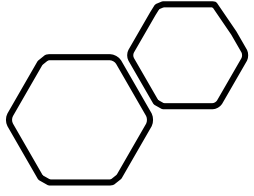
Data independence and efficient access.



Data integrity and security.



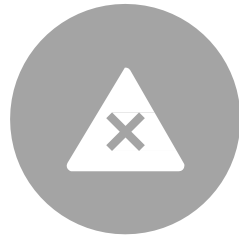
Concurrent access, recovery from crashes.



Benefits of Database System over File System



**REDUNDANCY CAN BE
REDUCED**



**INCONSISTENCY CAN
BE AVOIDED**



DATA CAN BE SHARED



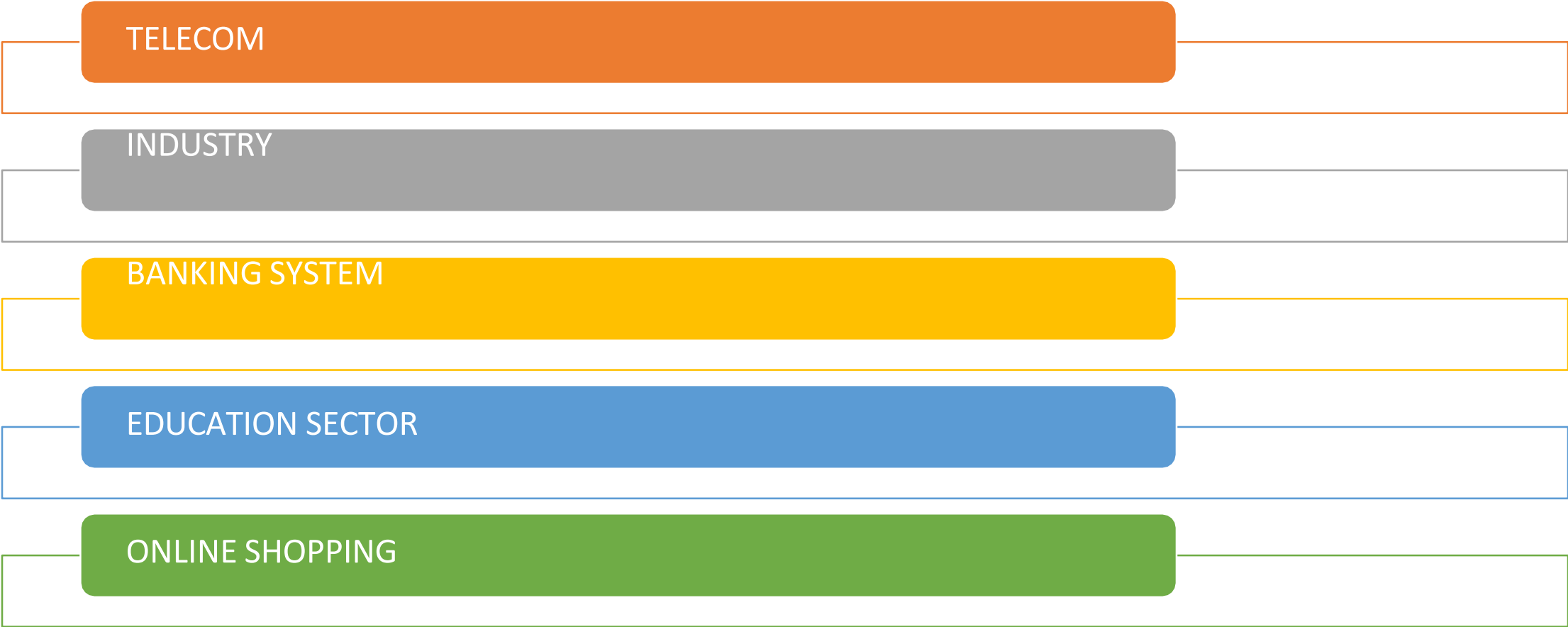
**SECURITY
RESTRICTIONS CAN BE
APPLIED**



**INTEGRITY CAN BE
MAINTAINED**

Applications of DBMS

TELECOM



The diagram illustrates five applications of Database Management Systems (DBMS). Each application is represented by a horizontal bar of a specific color, with the name of the application written in white capital letters inside the bar. To the left of each bar is a short vertical line of the same color, and to the right is a long horizontal line of the same color, forming a frame around the bar.

INDUSTRY

BANKING SYSTEM

EDUCATION SECTOR

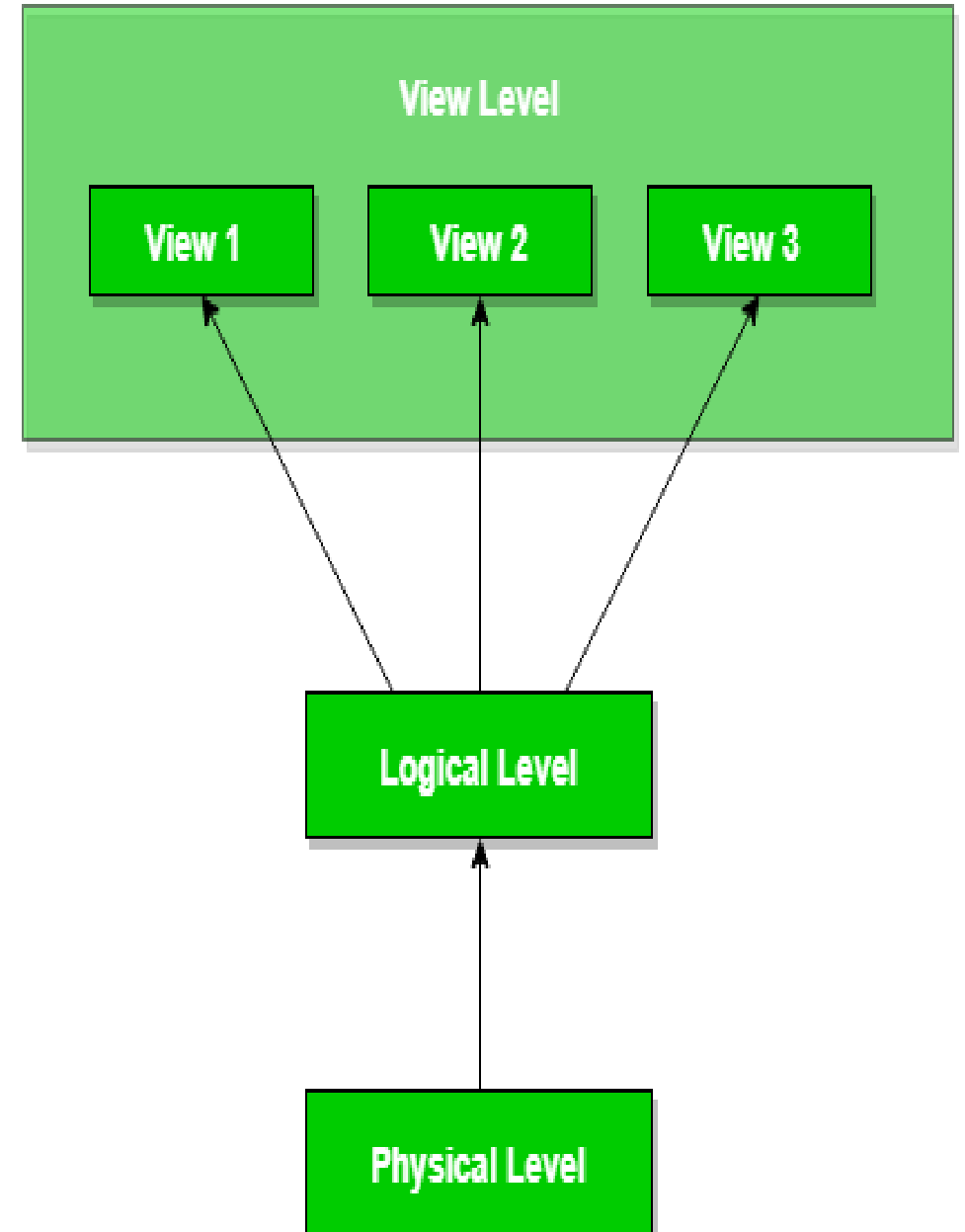
ONLINE SHOPPING

Three levels of Data Abstraction

Physical level: This is the lowest level of data abstraction.

Logical level: This is the middle level of 3-level data abstraction.

View level: Highest level of data abstraction.



Schema and Instance in DBMS

- Design of a database is called the schema
 - Physical Schema
 - Logical Schema
 - View Schema
- The data stored in database at a moment of time is called instance of database.



Relational Databases

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RDBs use Structured Query Language (SQL), which is a standard user application that provides an easy programming interface for database interaction.

Relational DBMS (RDBMS)

Introduction

Was introduced by Dr. E.F.Codd in 1970 .

Standard for relational databases is published by ANSI(the American National Standard Institute)

Features

A relational database is composed of many relations in the form of two-dimensional tables of rows and columns containing related tuples known as logical view.

Tables/Relations are a logical structure which is a collection of 2-dimensional tables consists of horizontal rows and vertical columns.



ADVANTAGES



USE OF OOPS
CONCEPT



HIGHLY
SECURED DATA.



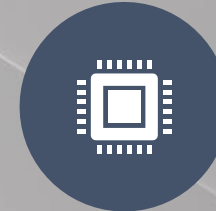
MULTIPLE
USERS CAN
ACCESS WHICH
IS NOT
POSSIBLE IN
SIMPLE DBMS.



SUPPORTS DATA
INDEPENDENCE.



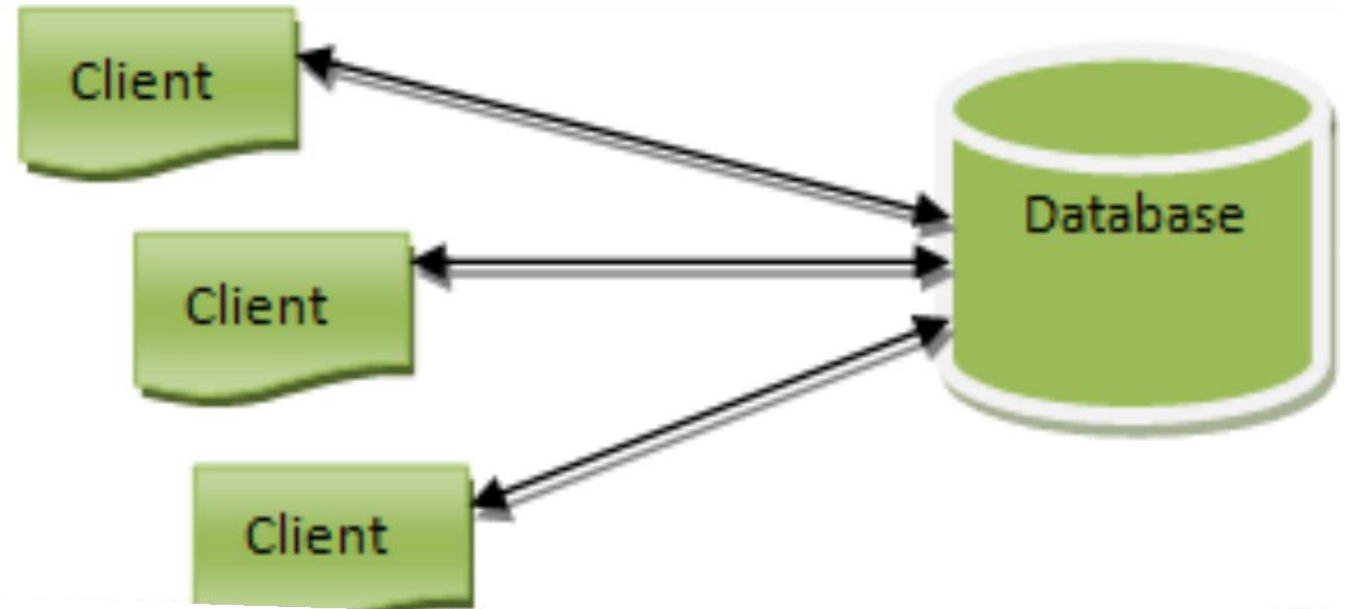
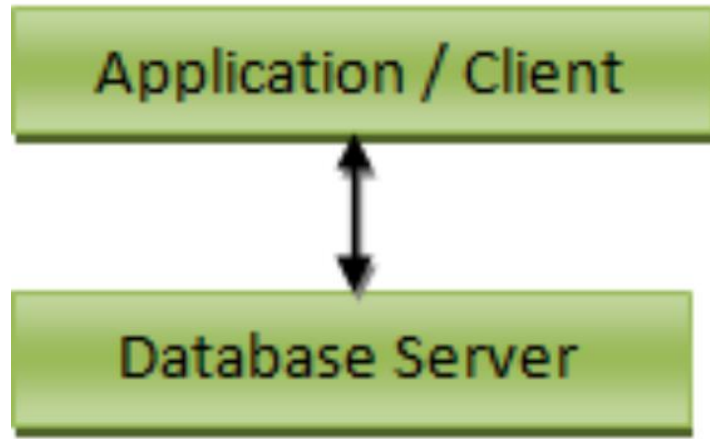
DISADVANTAGES



INABILITY TO HANDLE APPLICATION
AREAS LIKE SPATIAL DATABASES(EG-
CAD) ,APPLICATIONS INVOLVING
IMAGES ETC.

Database architecture

- Database architecture can be 2-tier or 3 tier architecture based on how users are connected to the database to get their request done. They can either directly connect to the database or their request is received by intermediary layer, which synthesizes the request and then it sends to database.



2-tier Database Architecture

- In 2-tier architecture, application program directly interacts with the database. There will not be any user interface, or the user involved with database interaction



Advantages of 2-tier Architecture:



Easy to understand



Easy to retrieve



Easy to modify



Easy to maintain



Disadvantages of 2-tier architecture:



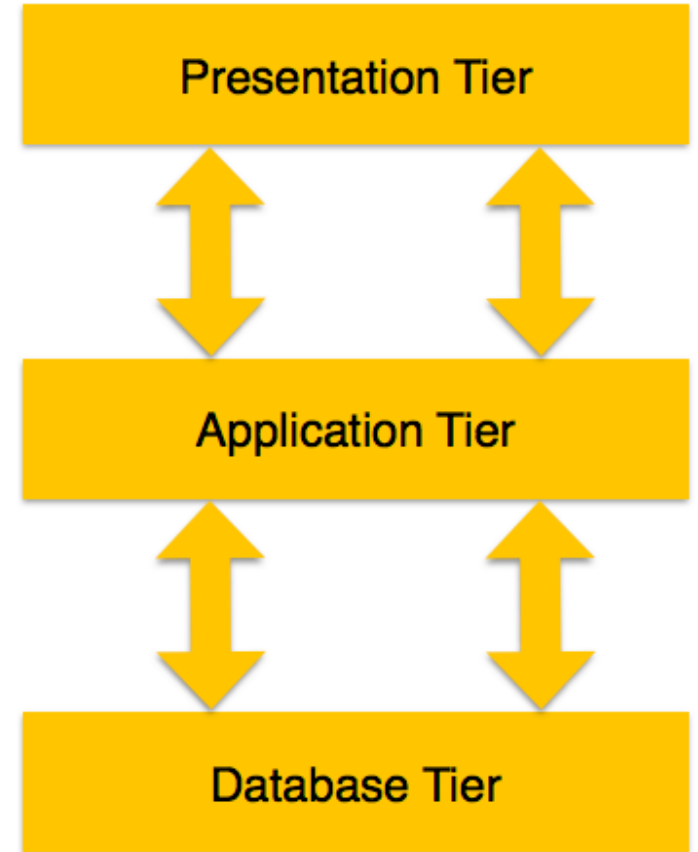
Time consuming



Cost effective.

3-tier Architecture

- A 3-tier architecture separates its tiers from each other based on the complexity of the users and how they use the data present in the database. It is the most widely used architecture to design a DBMS.



Database (Data) Tier – At this tier, the database resides along with its query processing languages. We also have the relations that define the data and their constraints at this level.

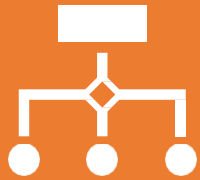


Application (Middle) Tier – At this tier the application server reside and the programs that access the database. For a user, this application tier presents an abstracted view of the database. End-users are unaware of any existence of the database beyond the application. At the other end, the database tier is not aware of any other user beyond the application tier. Hence, the application layer sits in the middle and acts as a mediator between the end-user and the database.



User (Presentation) Tier – End-users operate on this tier and they know nothing about any existence of the database beyond this layer. At this layer, multiple views of the database can be provided by the application. All views are generated by applications that reside in the application tier.

Transaction Management



A transaction is one or more SQL statements that make up a unit of work performed against the database, and either all the statements in a transaction are committed as a unit or all the statements are rolled back as a unit. This unit of work typically satisfies a user request and ensures data integrity.



For example, when you use a computer to transfer money from one bank account to another, the request involves a transaction: updating values stored in the database for both accounts. For a transaction to be completed and database changes to be made permanent, a transaction must be completed in its entirety.



Properties of Transaction



1. Atomicity



2. Consistency



Isolation



Durability



BEGIN TRANSACTION transfer

Deduct Rs.100 from account A

Add Rs.100 to account B

If no errors, then

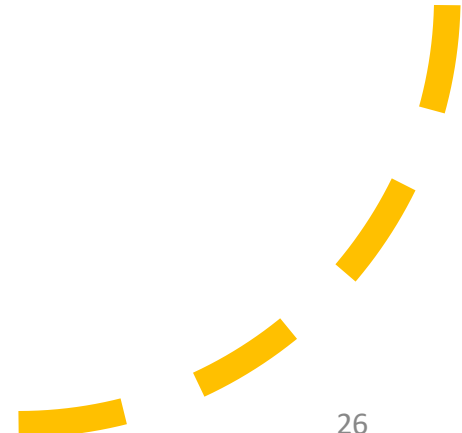
Save changes made by this operation

Else

Delete changes made by this operation

End If

END TRANSACTION transfer



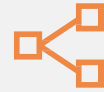
Chapter 2: Data Models



Data Model



Business Rules



Entity



Attribute



Relationships

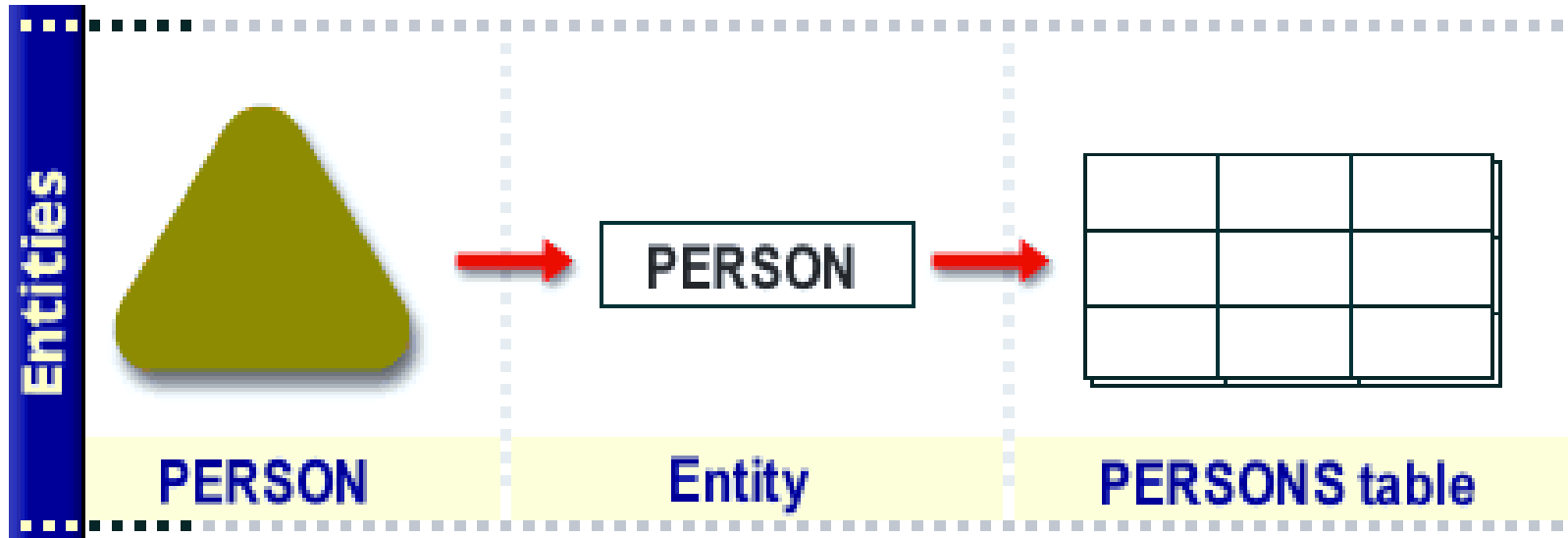
One to One

One to Many

Many to Many

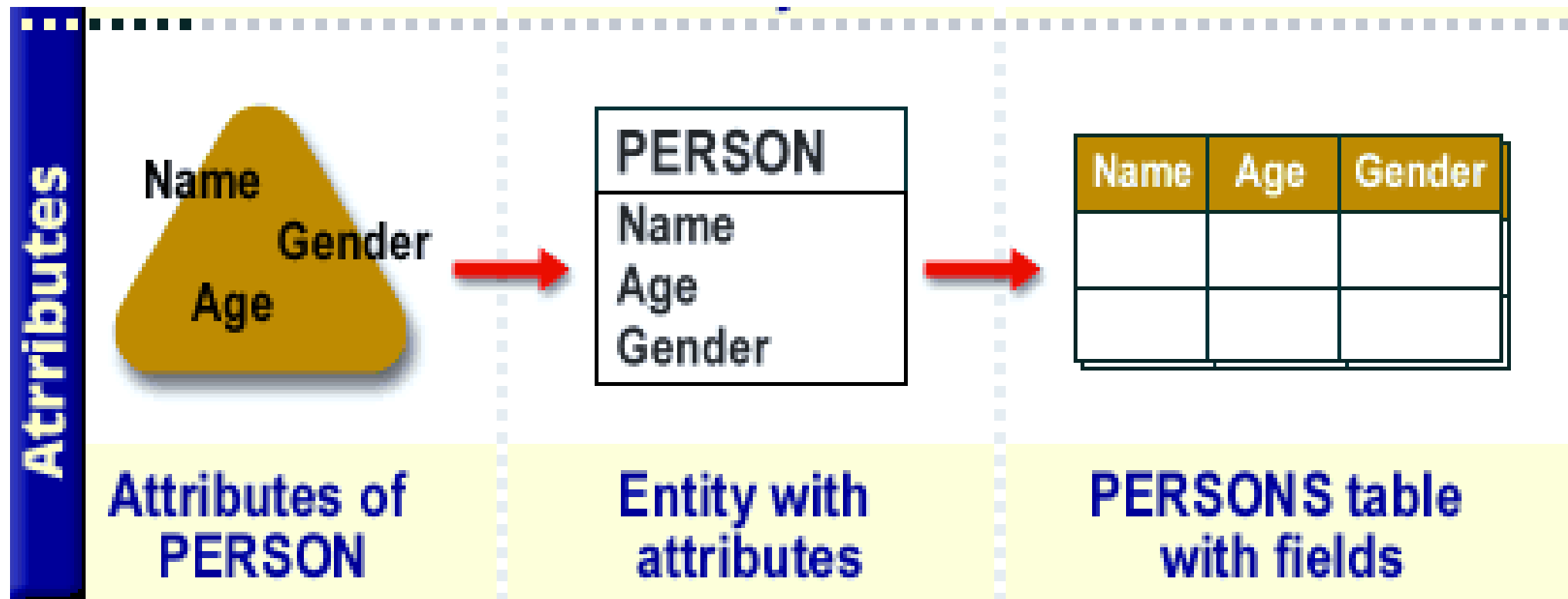
Entity

- A fundamental component of a model which has its own independent existence in real world. E.g. A Student, Faculty, Subject.
- An entity can be an object with physical existence, or it may have logical existence.
- Entities like department, section, adults may have physical or logical existence



Attributes

- Each entity has its own properties which describes that entity, such properties are called as attributes.
- A particular entity will have some value for each of its attributes.
- EXAMPLE – Employee entity may be described by attributes name, age, phone etc.



Relationship



It is an association among several entities for e. g. Employee works for Department.



The degree of the relationship is the number of participating entity types in a particular relation.



For example, there is a relationship between customers and agents that may be described as "an agent can serve many customers and each customer may be served by one agent".

Types Of Relationships



One is to one – One entity is associated with at most one other entity. E.g. One department can have only one manager



One is to Many – One entity is associated with any number of entities in other entity. E.g. One teacher may teach to many students



Many is to Many – Many entity is associated with any number of entities in other entity. E.g. Books in library issued by students



Entity Name

Entity

Person, place, object, event or concept about which data is to be maintained

Example: Car, Student



Jack

Attribute Name

Attribute

Property or characteristic of an entity

Example: Color of car Entity Name of Student Entity



Relation

Verb Phrase

Association between the instances of one or more entity types

Example: Blue Car Belongs to Student Jack

Business Rules



Database designer needs to take help from concepts such as entity, attributes and relationships to build a data model, but the above things are not sufficient to describe a system completely.



Business rules may define actors and prescribe how they should behave by setting constraints and help to manage business change in the system.



Business rules are statements of a discrete operational business policy or practice within specific organisations that constrains the business. It is intended to control or influence the behaviour of the business.



Examples of Business rules in Academia say University – Every student must have 75% attendance in order to sit for exam, A student has to pass 40 out of 100 in order to pass the subject, the maximum no of KT allowed per semester is 2 etc.

Types of Business Rules

- **DEFINITIONS** – Define some business terms. Definitions are incorporated in systems data dictionary. E.g. A professor is someone who teaches to students
- **FACTS** – Connect business terms in ways that make business sense. Facts are implemented as relationships between various data entities E.g. A professor may have student.
- **CONSTRAINTS** – Shows how business rules and how business terms are connected with each other. Constraints usually state how many of one data entity can be related to another data entity E.g. Each professor may teach up to four subjects.
- **DERIVATIONS** – Enable new knowledge or actions. Derivations are often implemented as formulas and triggers. E.g. A student pending fees is his fees paid minus total fees.

Characteristics of Business Rule

- **Atomicity** – Rule should define any one aspect of the system environment. E.g. College should have students in it.
- **Business Format** – Rule should be expressed in business terms understandable to businesspeople. E.g. ER diagrams, Object diagrams etc.
- **Business Commonality** – Each rule should confirm business scope. E.g. College system must define characteristics in college database.
- **Business Ownership** – Each rule is governed by a businessperson who is responsible for verifying it, enforcing it and monitoring need for change. E.g. End user or customer is responsible for requirements submitted by him.

- **Classification** – Each rule can be classified by its data and constraints.
- **Business Formalism** – Each rule can be implemented in the related information system. Business rules should be consistent and non-redundant.
- **EXAMPLES**
 - A student may take admission to college
 - One subject is taught by only one professor
 - A class consists of minimum 60 and maximum 80 students

Method Of Defining Business Rules



DEFINE THE
SCOPE OF
YOUR
ANALYSIS



STATE THE
OBJECTIVES
OF THE
DEFINED
SCOPE



IDENTIFY THE
ACTORS
WITHIN THE
DEFINED
SCOPE



UNCOVER
BUSINESS
CONFUSIONS



CREATE ONE
OR MORE USE
CASES



IDENTIFY
SYSTEM
DECISIONS

Hierarchical Model



First DBMS model



Data sorted hierarchically in top down or bottom up approach



Uses pointers to navigate between stored data



Represents data as a hierarchical tree

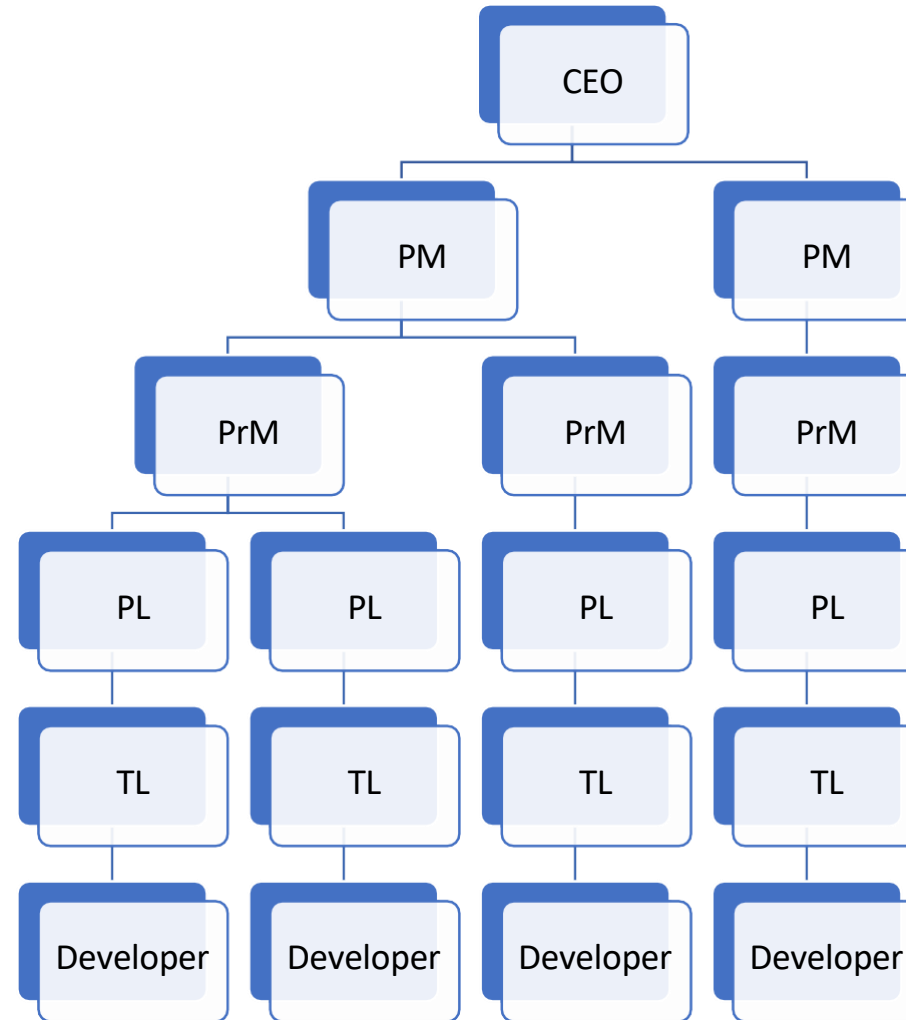


One parent node can have many child nodes but one child cannot have more than one parent.



EXAMPLE – Information Management System (IMS) from IBM

Example-Hierarchical Model



Advantages Of Hierarchical Model

Conceptual Simplicity

- Relationships between various levels is logically very simple. Hence database structure becomes easier to view

Database Security

- Security is given by DBMS itself it does not depends on programmer has given security or not

Simple Creation, Updation And Access

- It is simple to construct with the help of pointers. Adding and deleting records is easier . Faster and easy data retrieval

Database Integrity

- There is always parent child relation between different levels and hence child records are attached with parent records which maintains data integrity

Data Independence

- Itself maintains data independence thus reducing program efforts and its maintenance that is if one part of the code is changed no need to change the other part

Efficiency

- Good performance when database contains large amount of data in which one record has many related records

Disadvantages Of Hierarchical Model

Complex implementation

- Only data independence is not enough for designer and programmers to build database system they need to have knowledge of physical data storage which may be complex.

Difficult to manage

- Any change in location of data needs change in all application programs that accesses changed data. Data access is restricted by pointer path.

Lack of structural independence

- Change in database structure affects data access.

Complex application programming

- Programmers must know how data is stored and path of data storage.

Limitations in implementation

- 1:N relationship can be implemented but implementing M:N relationship is difficult.

Network Model

Uses pointers but without the need of parent child relationships.

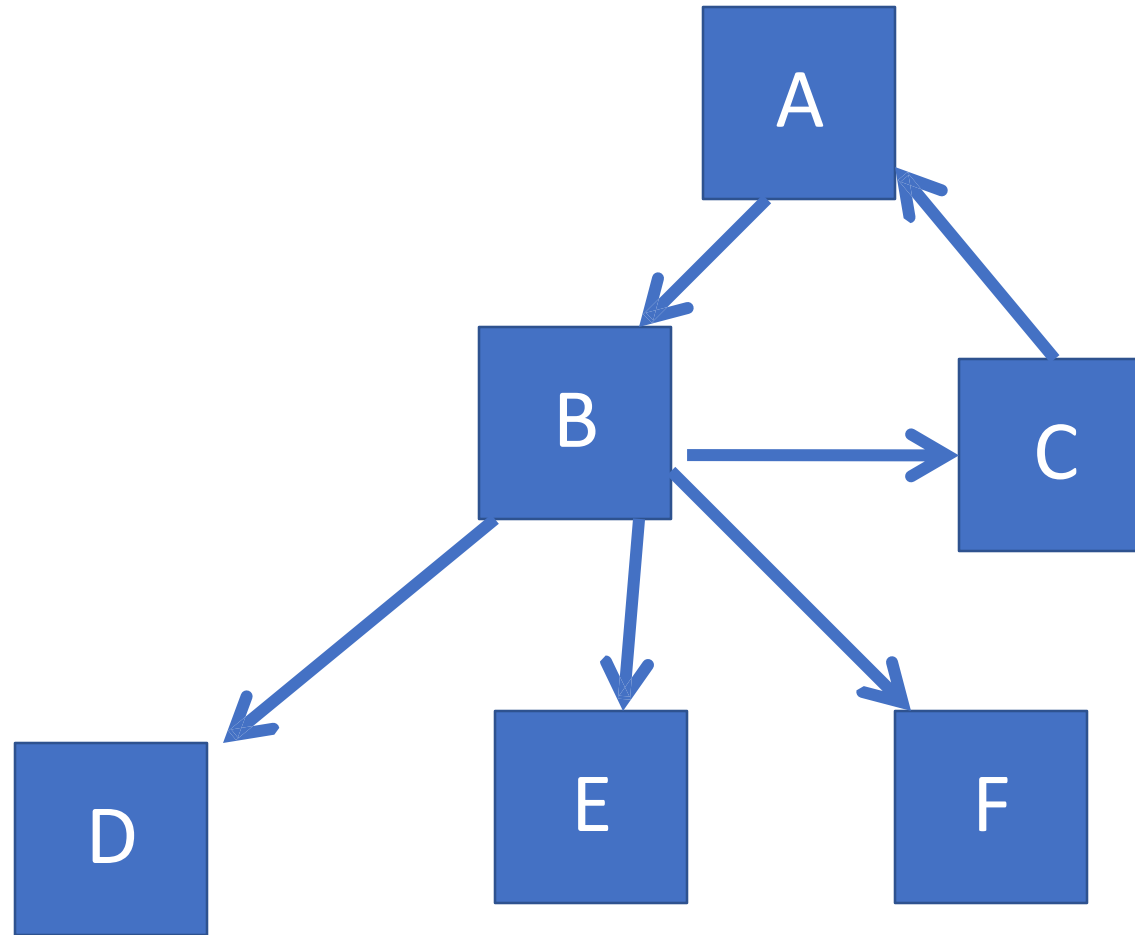
Used in network databases.

Relationship between any two record types is called as SET.

A relationship can be 1:N or M:N.

EXAMPLE – IDS(Integrated Data Store) one of the products based on network models developed by IBM and North American Rockwell

Example- Network Model



Advantages Of Network Model

Simple Design

- Is simple and easy to design and understand.

Ability to handle many types of relationship

- Can handle one-to-many or many-to-many or other relationships and hence manages multi user environment.

Ease of data access

- An application can access a parent record and all the member records within a set.

Data Independence

- Application programs work independently of the data. Any changes made in data do not affect the application program.

Conformance to standards

- Facilitates the administrators portability by offering data creation by DDL and DML.

Disadvantages Of Network Model

System Complexity

- Data are accessed one record at a time hence the complexity for the system increases for accessing multiple records at a time.

Lack Of Structural Independence

- Any changes made to the database structure or data require the application programs to be modified before it can access data.



Relational Model

Proposed by E. F. Codd

An attempt to simplify database structure by making use of tables and columns.

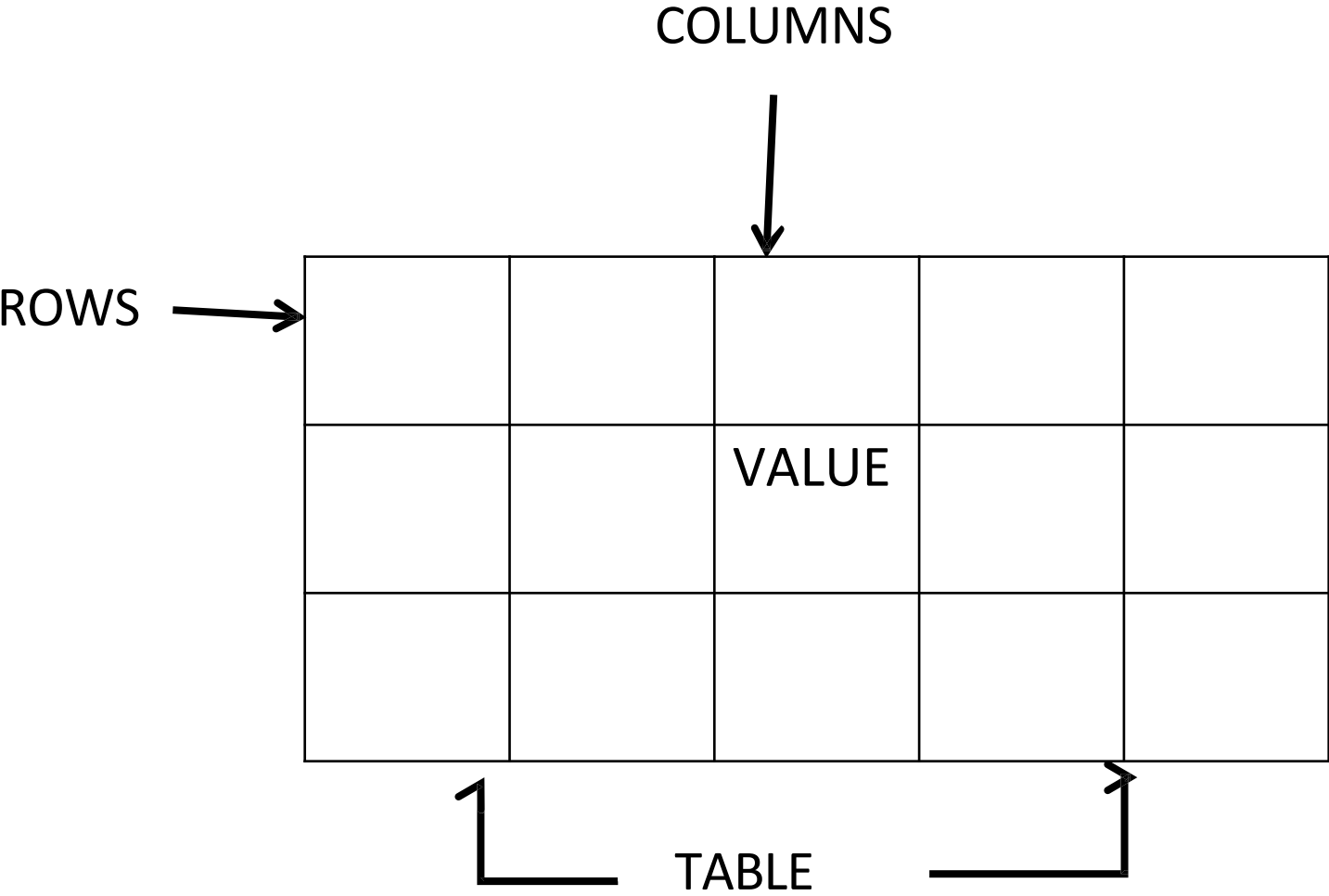
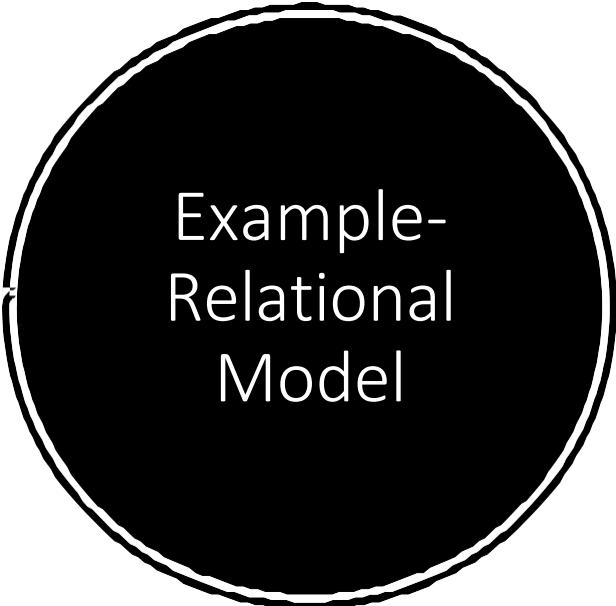
Collection of 2-dimensional tables which consists of rows and columns.

Tables are known as relations, columns are known as attributes and rows are known as tuples.

Uses collection of tables to represent relationships amongst data.

Each database item is viewed as a record with attributes. A record with similar attributes is called as table. Each table contains a record of particular type

The database uses relational model called as RDBMS (Relational Database Management System)



Advantages Of Relational Model

Relational Algebra

- Supports relational algebra, union, intersection, difference, Cartesian product, select, project, join and division

Dynamic View

- View is not a part of physical schema, hence changing data in the table also changes the data present in the view

SQL

- English like language which can be used to access data from RDBMS. Most vendors support SQL

Excellent Data Security

- Support the concept of user rights thus meeting the needs of databases

Performance

- Performance and support to new hardware technologies and flexible for all types of data needs

Scalability

- Scalable and provide good support for the implementation of distributed systems and other advanced database systems

Disadvantages Of Relational Model

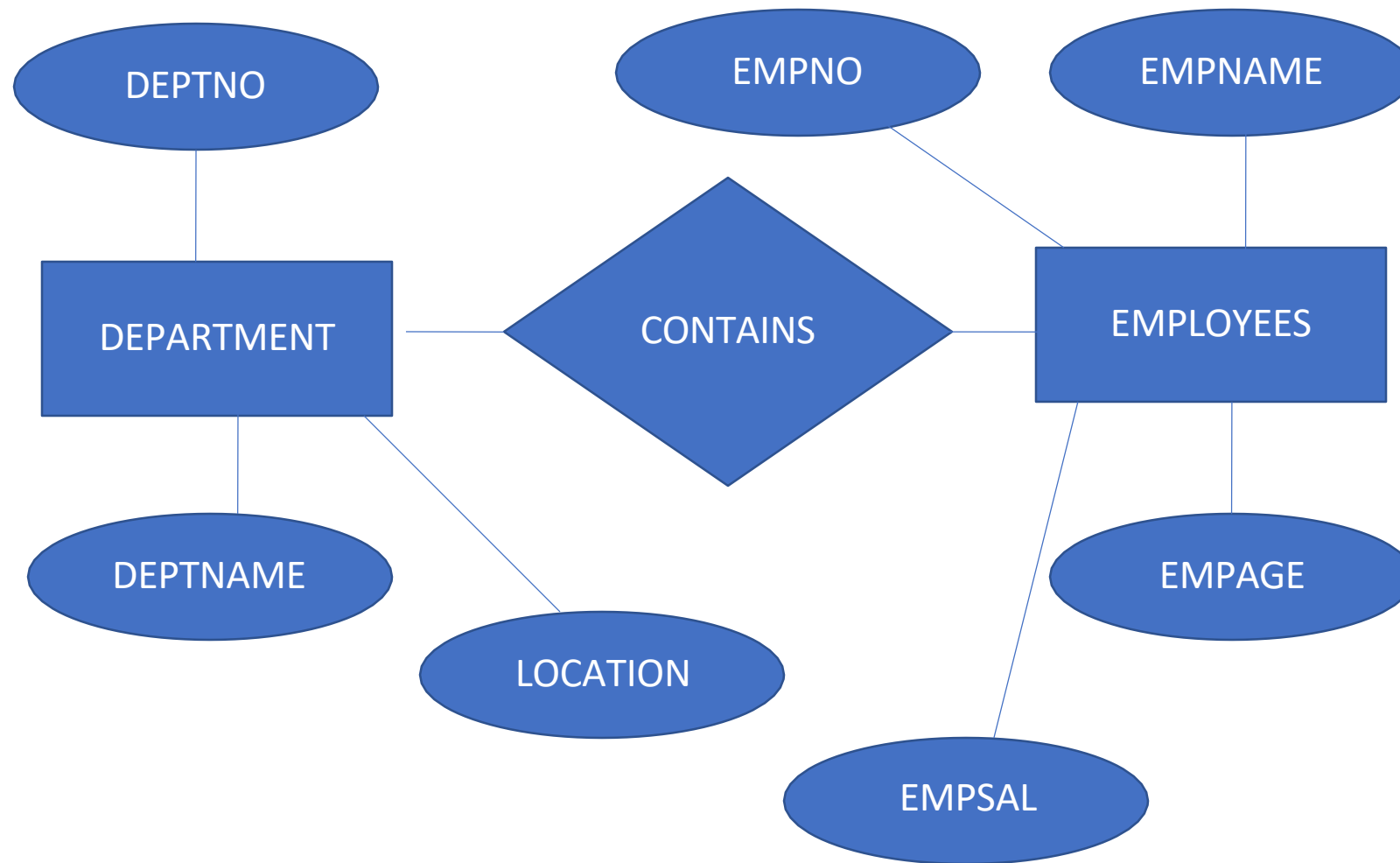
Relational databases can sometimes become complex as the amount of data grows, and the relations between pieces of data become more complicated.

Complex relational database systems may lead to isolated databases where the information cannot be shared from one system to another.

Entity Relationship Model

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- Based on the perception of real-world data that consists of set of entities (data items) and relationships among these entities.
- Popular high-level conceptual model used for conceptual design of database.
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- Relationship is an association among several entities and is illustrated by diamond and read from left to right.

Entity Relationship Model Contd..



Advantages Of ER Model

Very Simple Design

- Simple and easy to design logical view of data.

Ease Of Representation

- Designer, programmer, developer can understand system very easily by looking at an ER model constructed.

Integrated With Relational Databases

- Completely integrated with relational model which can offer well structured design process.

Disadvantages Of ER Model

Limited Constraint Representation

- Model can display those constraint that are directly tied by relationships but not possible to represent some constraints like average marks etc.

No DML

- Do not have any language by which we can insert data in database.

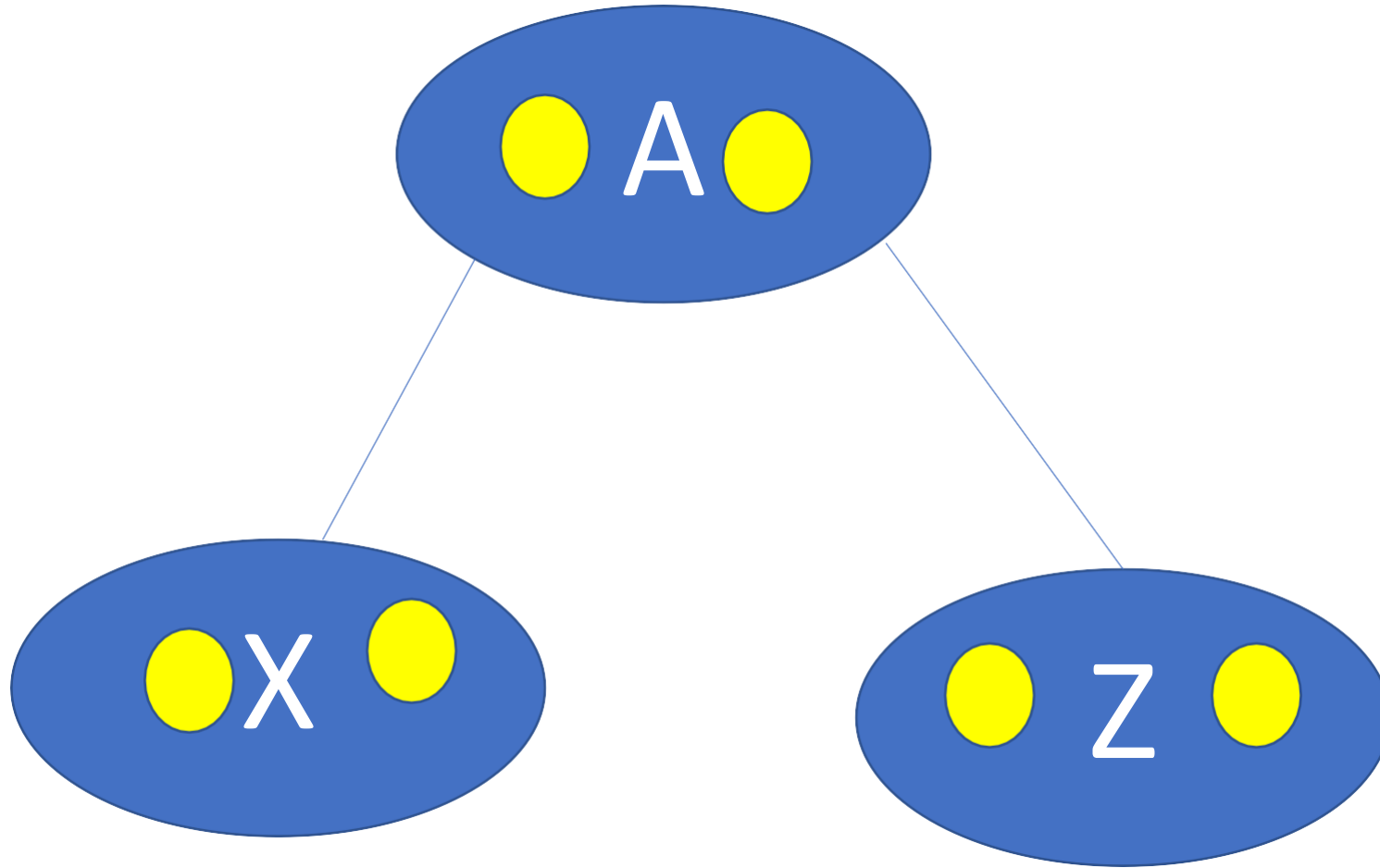
Loss Of Information Content

- Does not deal with actual data values so it may not be possible to represent information content of system.

Object Model

- Data is stored in the form of objects which are structures called classes that display the data within it.
- Fields are instances of these classes called objects.
- Used in File Management System.
- DBMS developed is called OODBMS (Object Oriented Database Management System).
- OOD can handle complex applications such as scientific experiments, geographical information system etc.
- Draws its concept from real world objects which deals with data at higher level that is with the objects surrounding the data.
- Represents database in terms of objects, attributes and their behaviors.

Object Model Contd..



Advantages Of Object Model

OO features provide a clear modular structure which is good for designing abstract data types where internal implementation details are hidden.

Easy to maintain and modify existing code as we can create new model with small change in existing.

Disadvantages Of Object Model

- Often provided through object-oriented languages such as C++ and Java.
- Practically very complex and inapplicable many times.

Evolution of Data Models

GENERATION	TIME	MODEL	COMMENTS
First	1960s-1970s	File System	Used mainly on IBM mainframe system Managed records ,not relationships
Second	1970s	Hierarchical and network	Early database systems Navigational access
Third	Mid-1970s to present	Object-oriented Extended Relational	Support complex data Extended relational products support Objects and data warehousing
Next Generation	Present to future	XML	Organization and management of unstructured data. Relational and object models add support for XML documents

Goals of Database Design

01

To full fill information requirements of the users and application.

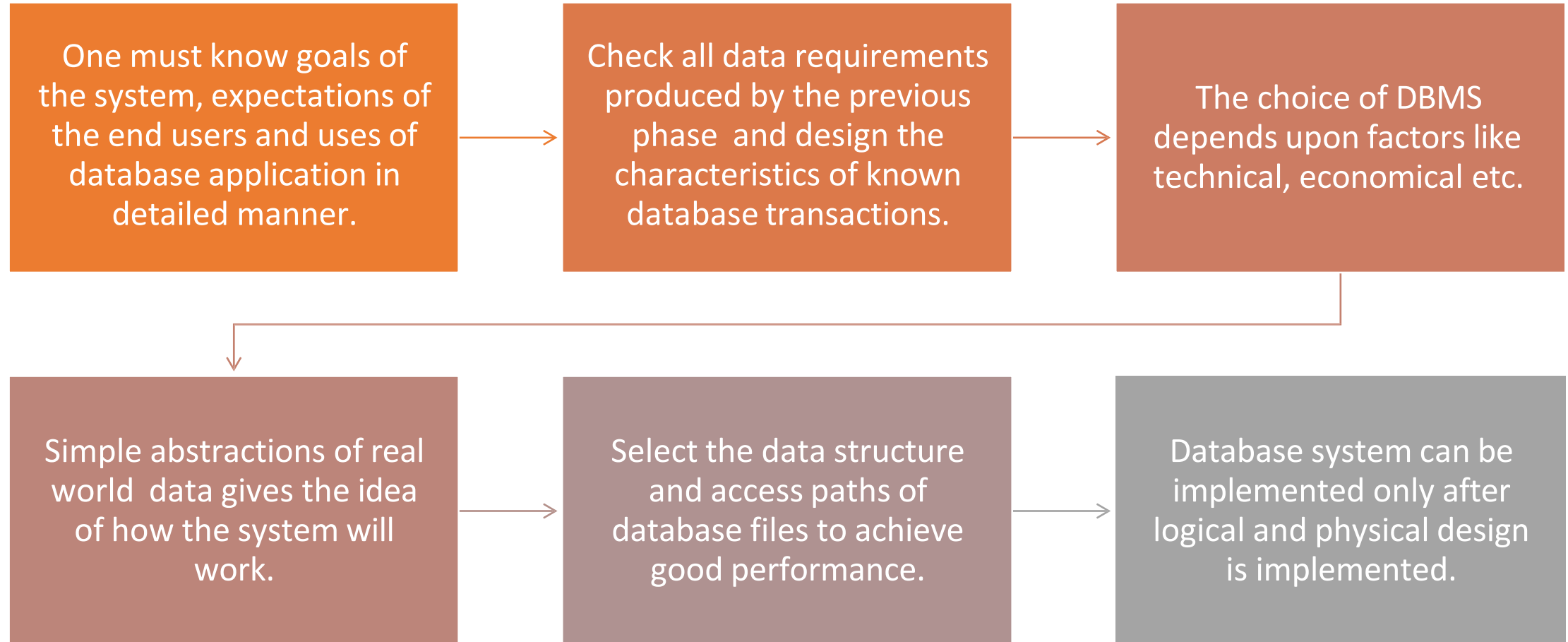
02

To support all processing requirements as well as all performance criteria like response time, processing time and storage space.

03

Provide very simple and easy to understand structure of the information.

Phases of database design process



Overview of ER Model

In 1976 Chen developed the Entity Relationship (ER) model a high level data model that is useful in developing a conceptual design for a database.

Constructing ER diagram is a first step in designing a database.

ER model defines the data elements and relationships among them.

ER data model is based on perception of real world data consisting of entities (data items) and relationships among those entities.

Popular high level conceptual model used for conceptual design of database.

ER Diagrams

Diagrams/Notations used in ER diagrams

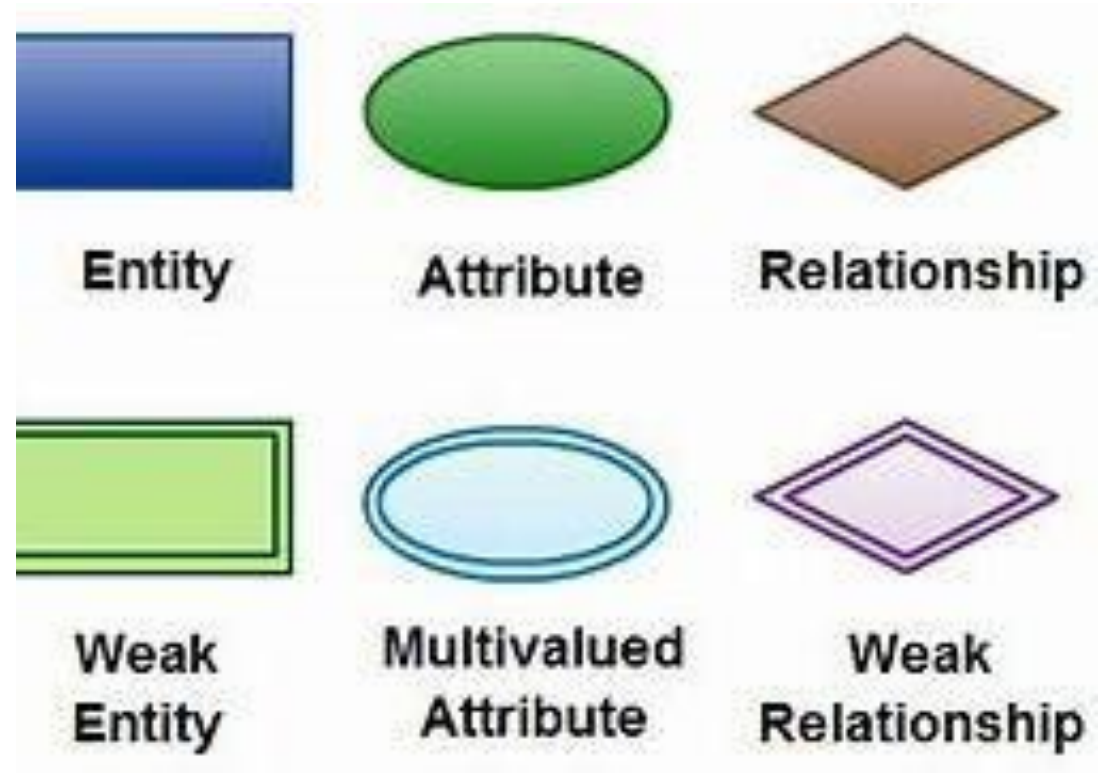
Rectangle entity sets

Ellipses attributes

Diamonds relationship sets

Lines link attributes to entities and entities with entities

Double Rectangles weak entity sets



Entity

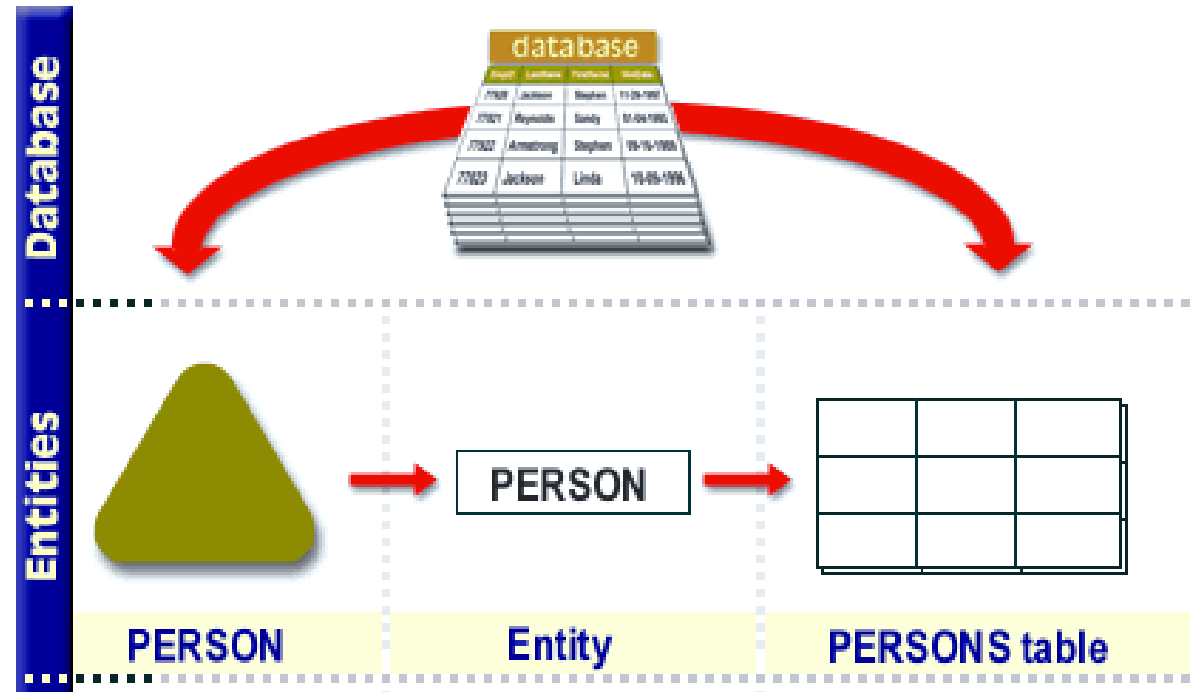
A fundamental component of ER model

It is a thing in real world with its own independent existence.

E.g. Student, Faculty

May be an object with physical or logical existence

Has its own properties that describe the entity known as attributes



Entity set

Collection of all entities of same type

Strong Entity –

- Entity type having own distinct primary key by which we can identify specific entity uniquely.
- E.g. Empno in Emp table, RollNo in Student table. Represented by single rectangle

Weak Entity –

- Entity type which cannot form distinct primary key from their attributes.
- These type of entities are dependent on strong entity for primary key.
- Some weak entities contain virtual primary key called as Discriminator. Represented by double rectangle



Strong entity set	Weak entity set
A single rectangle is used for the representation of a strong entity set.	A double rectangle is used for the representation of a weak entity set.
It contains sufficient attributes to form its primary key.	It does not contain sufficient attributes to form its primary key.
A diamond symbol is used for the representation of the relationship that exists between the two strong entity sets.	A double diamond symbol is used for the representation of the identifying relationship that exists between the strong and weak entity set.
A single line is used for the representation of the connection between the strong entity set and the relationship.	A double line is used for the representation of the connection between the weak entity set and the relationship set.
Total participation may or may not exist in the relationship.	Total participation always exists in the identifying relationship.

Attributes



Various properties that describe an entity



Attribute value that describes each entity becomes a major part of data stored in database as each entity will have some value for each of its attributes.



E.g. Employee entity has name, age, phone etc. as attributes

Attributes

Simple Attributes

Composite Attributes

Single Valued Attribute

Multi Valued Attribute

Stored Attributes

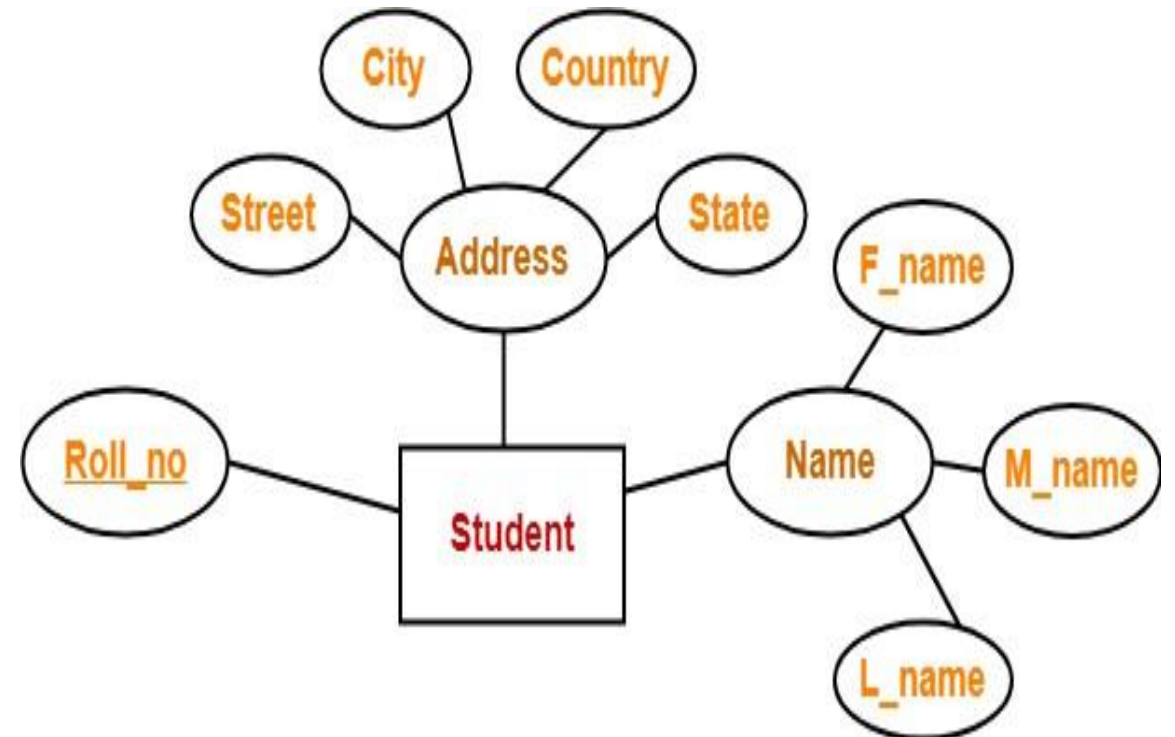
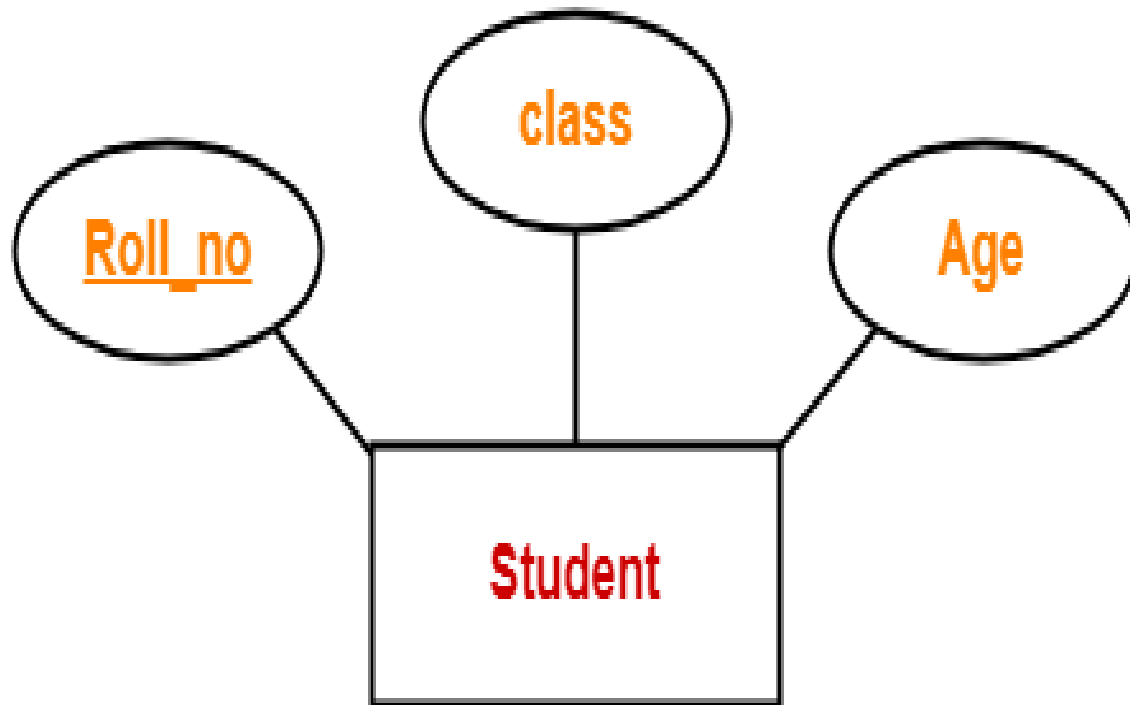
Derived Attributes

Null Attribute

Key Attribute

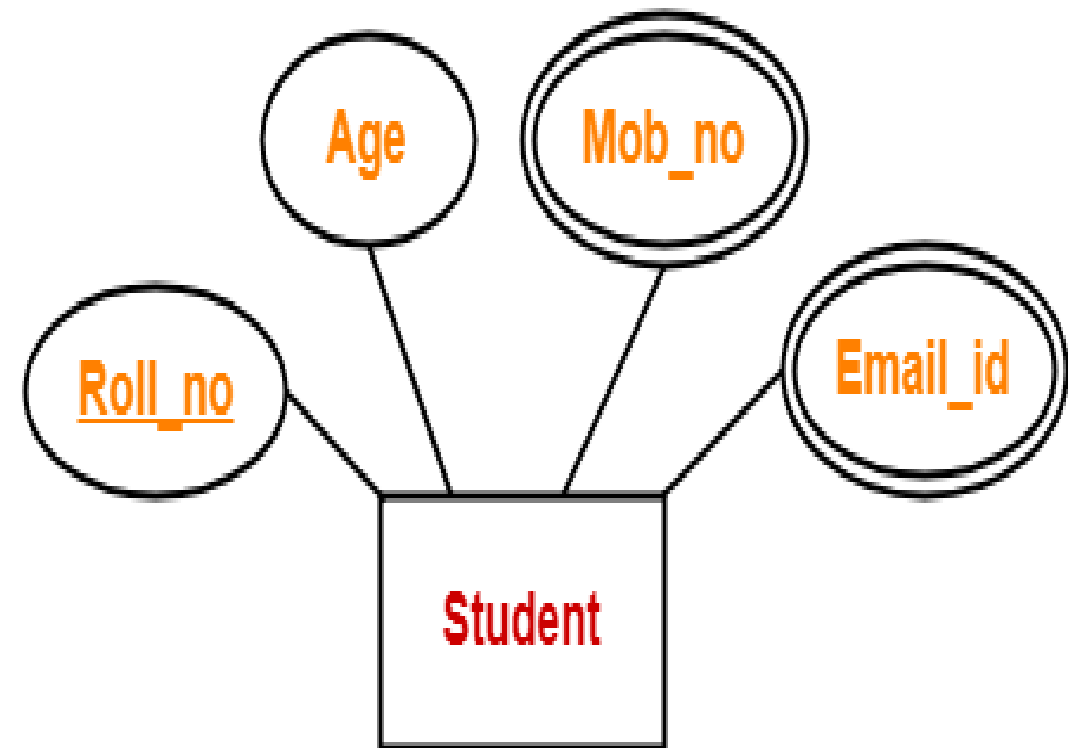
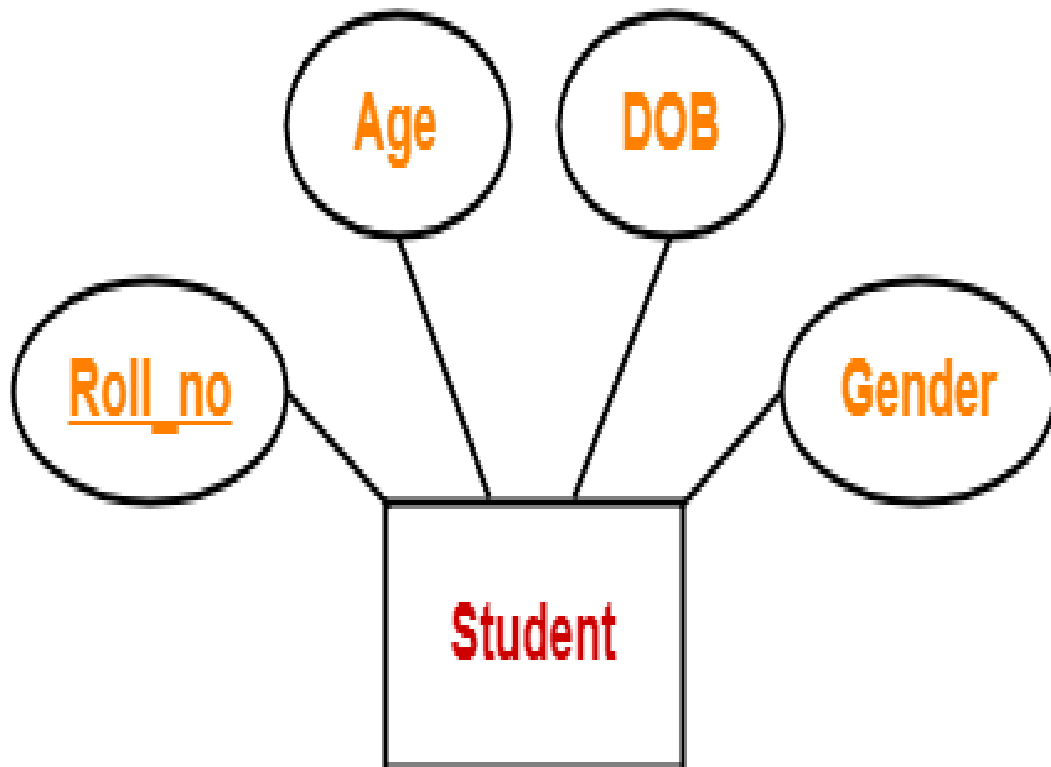
Attributes

- Simple Attributes: Which cannot be divided into sub parts.
Example: Salary of employee
- Composite Attributes: Which can be divided into sub parts E.g.
Name can be divided into FirstName and LastName



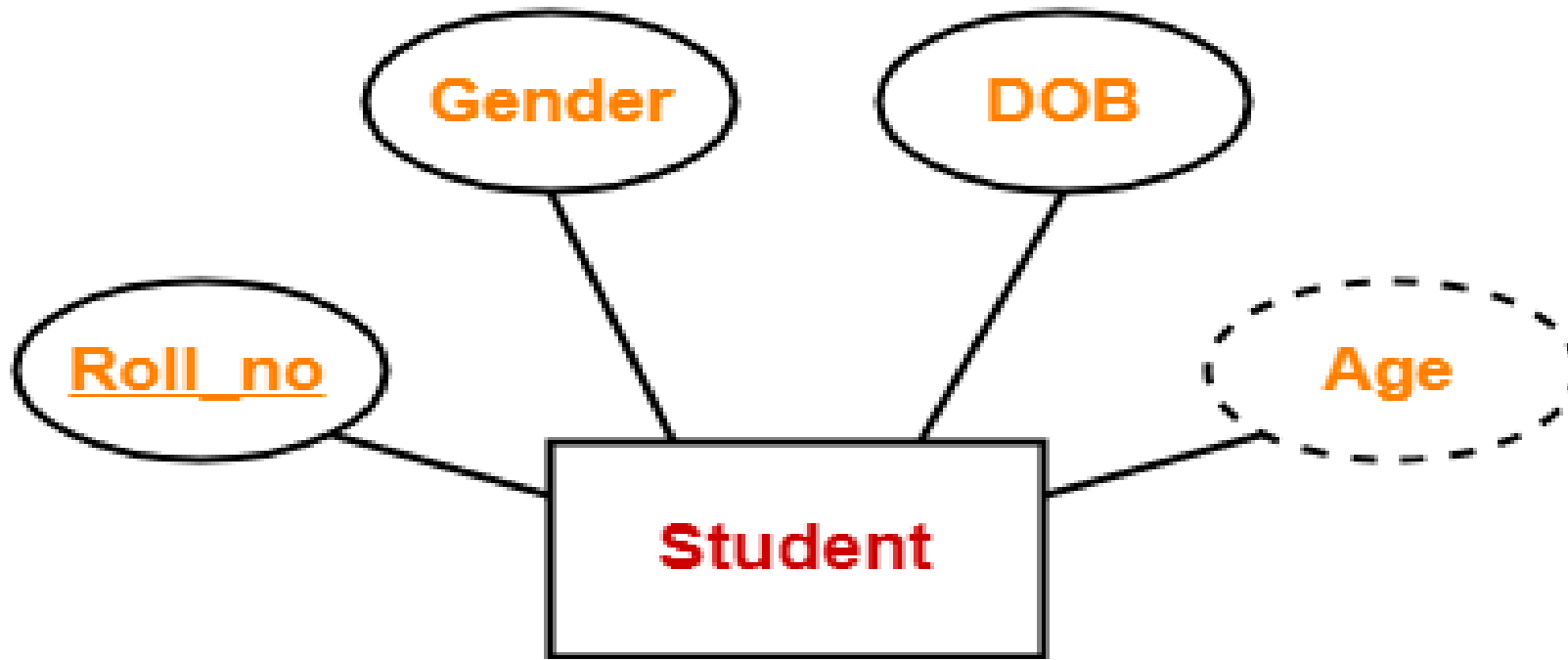
Attributes

- Single Valued Attribute: Having atomic (only one) value for particular entity. E.g. Student has only one roll no
- Multi Valued Attribute: Many values for a particular entity E.g. Employee has multiple mobile numbers



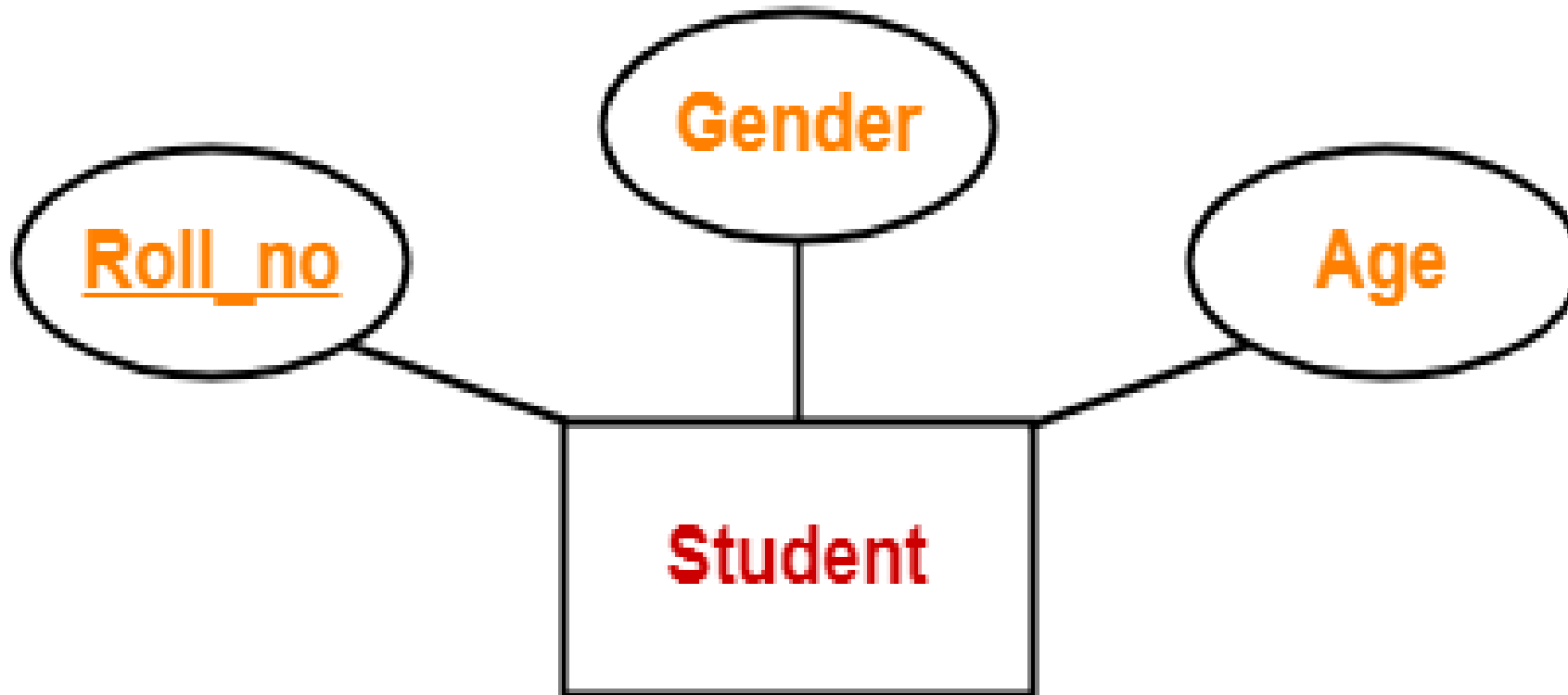
Attributes

- Stored Attributes: Simple attributes stored in database
E.g. DateOfJoin for Employee
- Derived Attributes: Value of this attribute is derived from value of related stored attribute E.g. EmployeeTenure can be calculated from DateOfJoin



Attributes

- Null Attribute: Can take a null value when entity does not have a value for it or the value is unknown. E.g. Commission attribute in Employee table specifies whether the Employee has commission or not
- Key Attribute: Must have a unique value by which any row can be identified. E.g. Deptno for department table



Constraints On Relationships

Mapping Constraints/ Cardinalities

ONE is to ONE – One tuple in entity is related only with one tuple in another entity. One department can have only one manager

ONE is to MANY – One tuple in entity is related with many tuples in another entity. One department can have many employees

MANY is to MANY – Many tuples in entity is related many tuples in another entity. Books in library issued by students

Constraints On Relationships

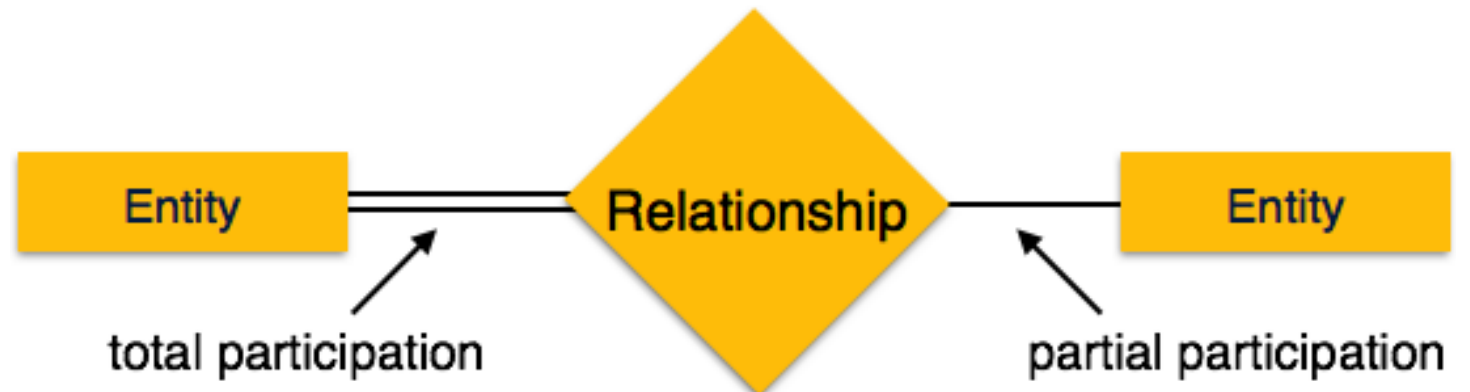
Participation Constraints

Total Participation

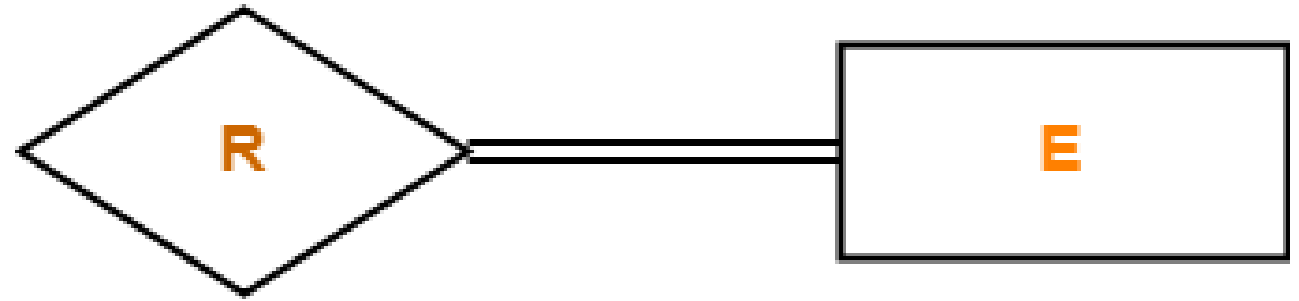
- Every object in an entity must participate in relationship
- Indicated by dark or double line between entity and relationship
- Every department must have manager

Partial Participation

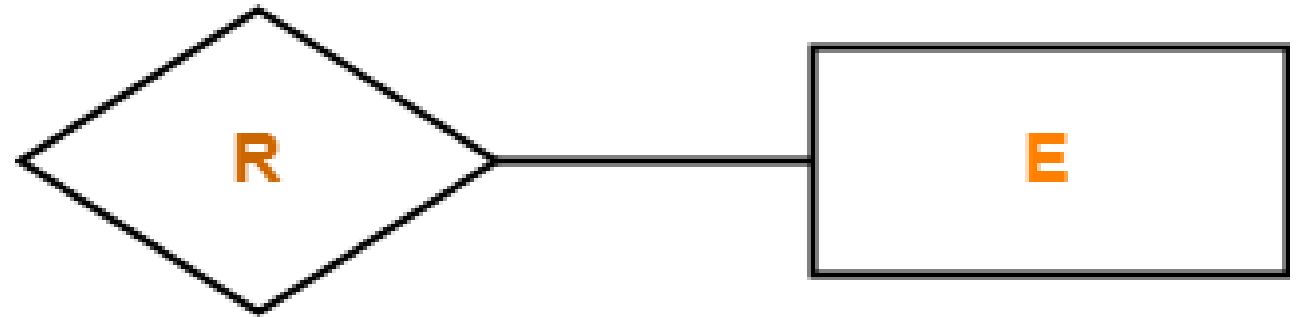
- More than one object in an entity may participate in a relationship
- Indicated by single line between entity and relationship
- Employee works for department



Constraints on Relationships



Total Participation



Partial Participation



Notations Used in ER Diagrams

Cardinality Ratio Notation

- Use cardinality of each identity involved in relation
- 1 (Exactly One Instance), M (Many Instances), N (Any Number Of Instances) specifies on each entity of binary relationship

Min-Max Notation

- (Min , Max) notation represents entity is related to at least min, at most max relationship instance in relationship set

Limitations Of Entity Relationship Model

One cannot represent relationship like inheritance in ER diagrams

No relationships is defined between entity type and relationship type

No relationship is defined between entity type and collection of entity type from which one may participate.

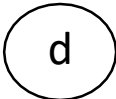
Extended Entity Relationship Model – ERD Issues



- EER model includes all modeling concept of ER model
- In addition it includes the concept of specialization and generalization
- EER model is used to represent collection of objects that is union of objects of different entity types
- A diagrammatic technique for displaying the EER concepts is called as EER diagrams

Sub Class – Super Class And Inheritance



- Super Class
 - The class has its sub groupings
 - An entity cannot exist in database merely by being member of any super class
- Sub Class
 - A sub grouping of super class
 - More specific version of super class
 - Inherits properties and attributes from its super class
- Super Class And Sub Class Relationship
 - Relationship between super class and sub class is called Superclass/Subclass relationship
 - Shown by encircled 
 - Leads to concept of Inheritance

Inheritance

- Inheritance
 - Important concept associated with subclass
 - Type of an entity is defined by an attribute poses by a relationship types in which they participates
 - Entity in subclass represent entity from upper class it should poses value for specific attribute as member of super class
 - Entity represents all relationships in which super class participates
 - Entity that is the member of subclass inhabits all attributes of entity as a member of super class

Specialization And Generalization

- Specialization
 - Top down approach of superclass/ subclass relationship
 - Process of defining a set of subclass of entity type called also as super class of specialization
 - Set of subclasses that forms a specialization is defined on the basis of some distinguishing characteristic of entity in super class
 - E.g. Set of subclass (Saving_Account, Current_Account) are specialization of superclass Account
 - The subclass defined in specialization is attached by lines to a circle which is connected to super class
 - Subset symbol on each line connecting a subclass to circle indicates the direction of super class/ subclass relationship
 - An attribute applied only to entities of particular subclass is called as specific attribute

Generalization



Reverse process of specialization or bottom up approach of super class/ subclass relationship



Process in which we differentiate among several entity types identifying their common features and generalizing them into single super class of which original entity type are special subclasses



E.g. Car and Bike do have several common attributes that can be generalized to super class vehicle



In diagrammatical notation arrow pointing to generalized super class represents generalization and arrow pointing to generalized subclass represents specialization



Attributes created of higher or lower level entities are attributes inheritance

Codd's Rule

- Information Rule – All available data in system should be represented as relations or tables.
- Guaranteed Access Rule – Each data item must be accessible without ambiguity by providing table name and its primary key of the row also include its column name to be accessed
- Systematic Treatment Of Null Values – Null values are not equal to blank space or zero they are unknown unassigned values which should be treated properly
- Self Describing Database – There should be dynamic online catalog based dictionary on relational model which keep information about tables data in database

Codd's Rule

- Comprehensive Data Sublanguage – The data access language (SQL) must be the only means of accessing data stored in the database and support DML, DDL etc.
- View Updating Rule – All views of data are theoretically updateable can be updated using system also
- High Level Insert, Update And Delete – This rule states that in a relational database , the query language must be capable of performing manipulations on sets of rows in a table
- Physical Data Independence – Any changes made in the way is physically stored must not affect applications that access data

Codd's Rule

- Logical Data Independence – This rule states that changes to the database to the database design should be done in a way without the users being aware of it
- Integrity Independence – Data integrity constraints which are definable in the language must be stored in the database as data in table is, in the catalog and not in the application program
- Distribution Independence – In a RDBMS data can be stored centrally that is on a system or distributed across multiple systems
- Non Subversion Rule – This rule states that there should be no bypass of constraints by any other languages

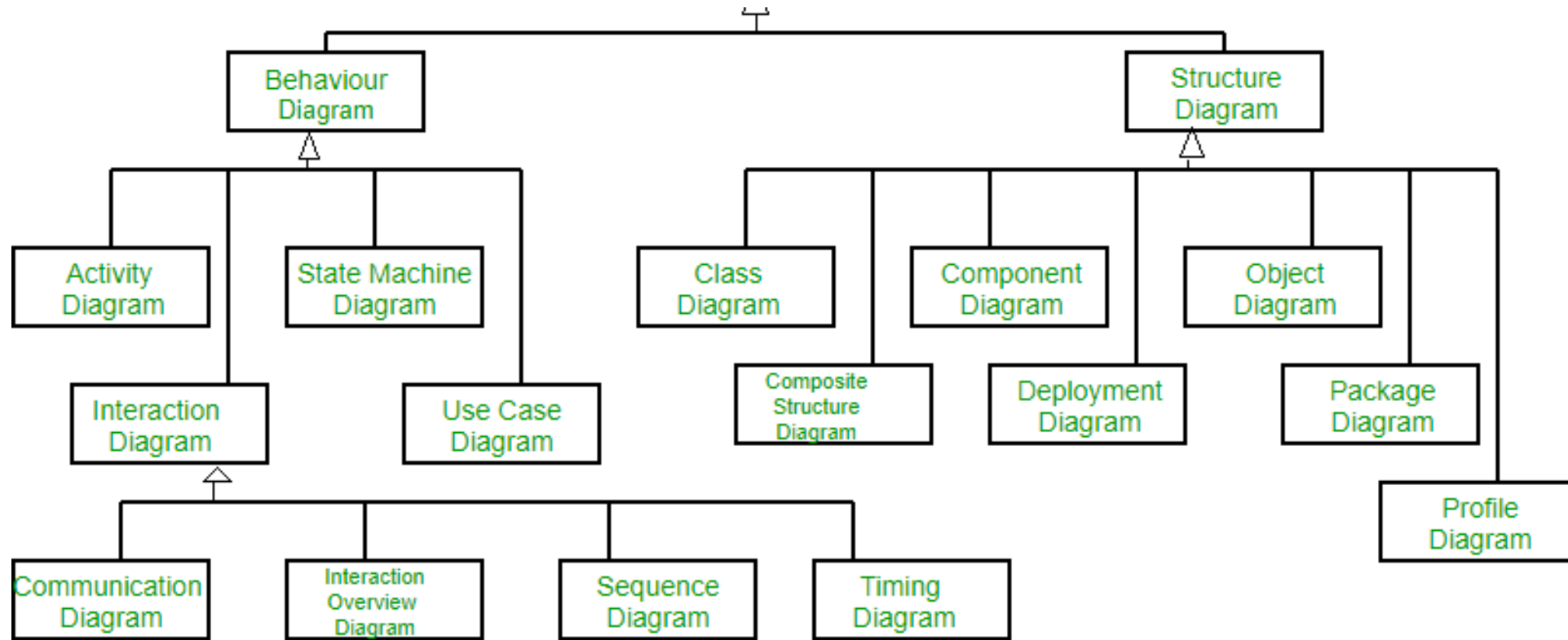
UML(Unified Modelling Language)

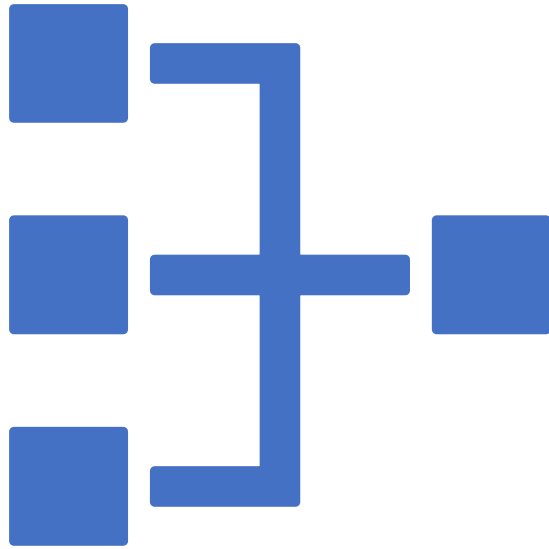
- Introduction of UML
- Class Diagram
- Use-case Diagram
- Activity Diagram
- Implementation Diagram
- Advantages and Disadvantages

Introduction To UML

- UML is a standard language for specifying visualizing, constructing and documenting the artifacts of software systems.
- UML is specially proposed standard for creating specifications of various components of a complex software system.
- UML stands for Unified Modelling Language.
- UML is different from the other common programming languages like C++,Java etc.
- UML is a specification language that is used in the software engineering field.

Types Of UML Diagrams

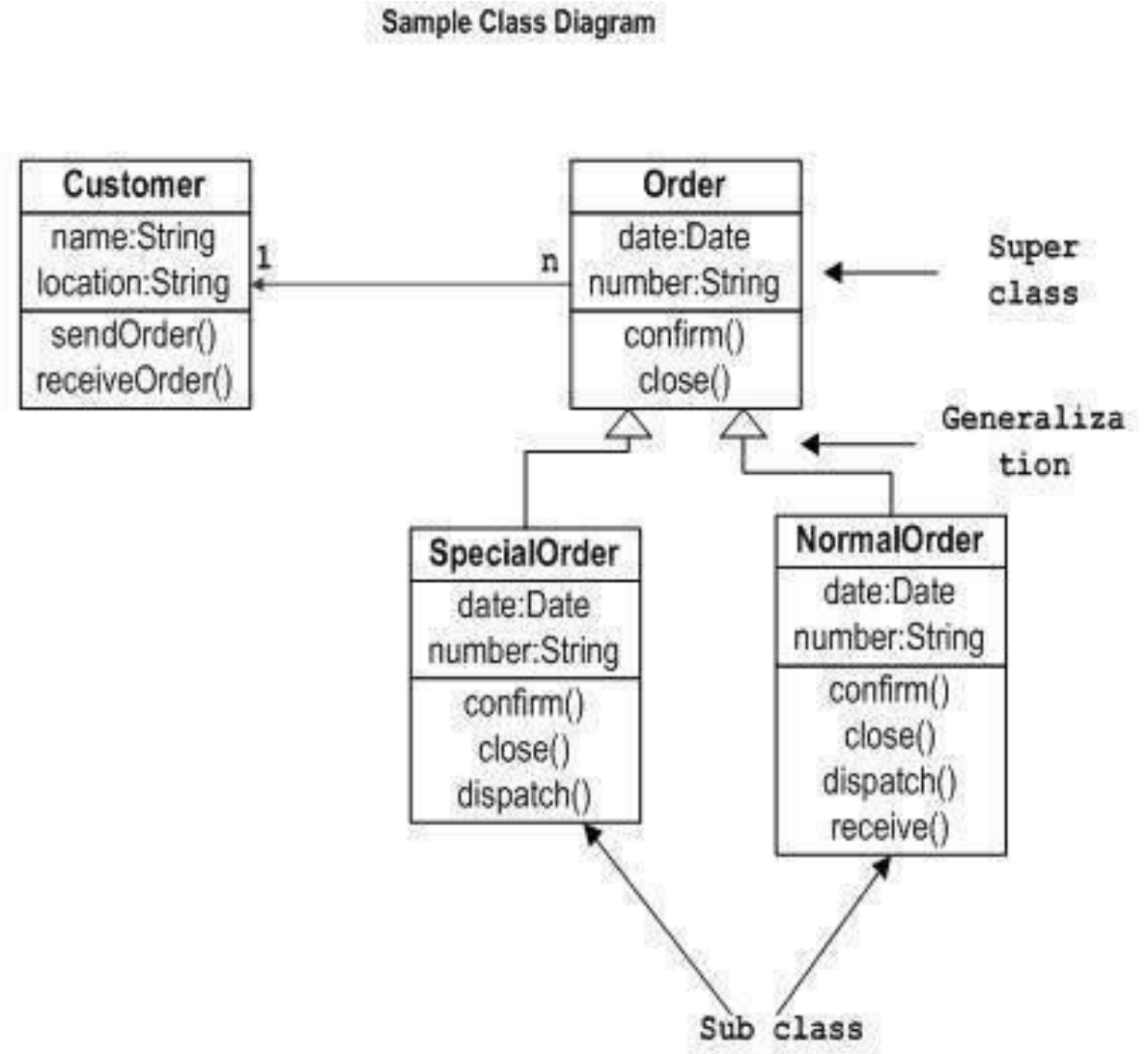




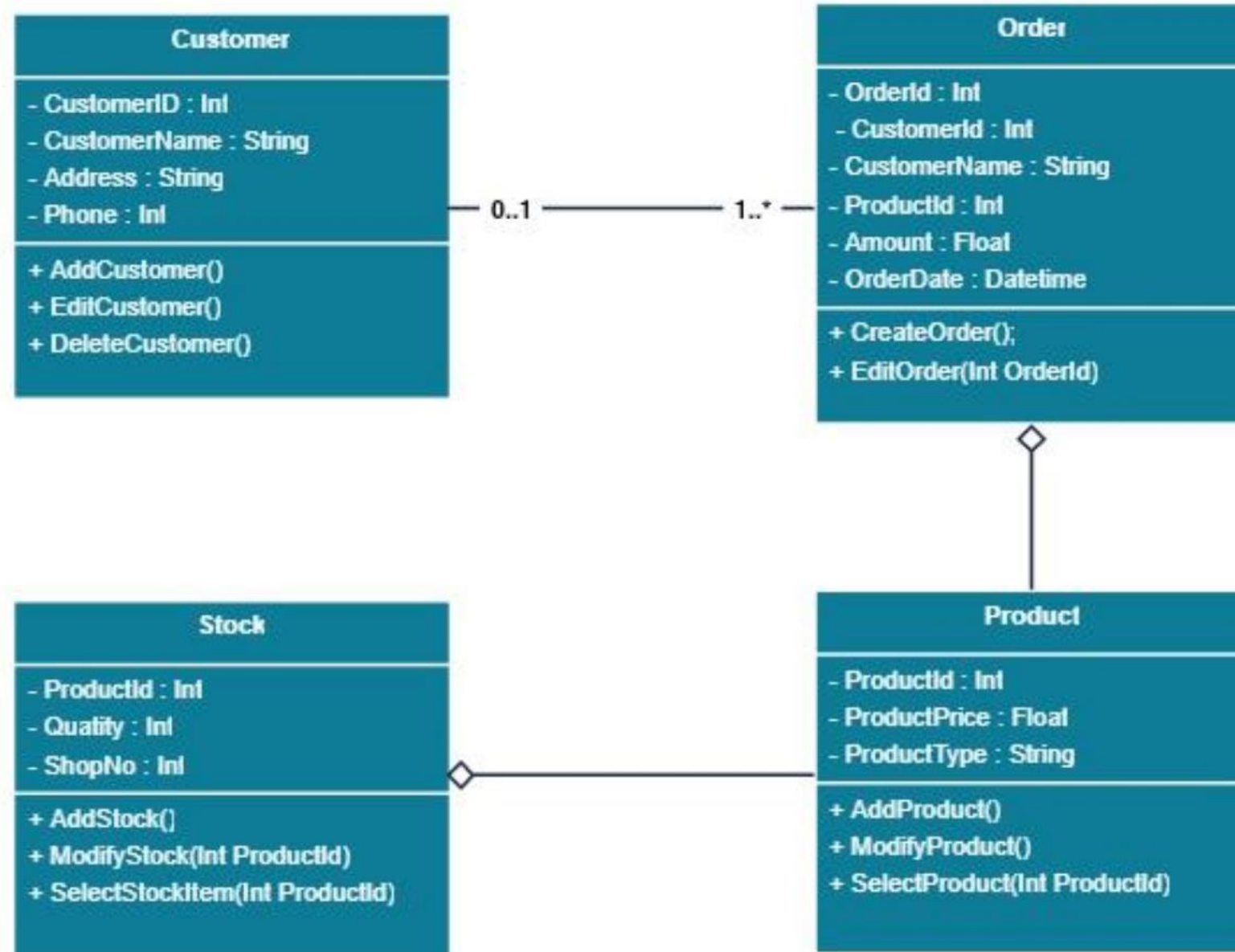
Class diagram

- Most popular UML diagrams used for construction of software applications.
- Similar to E-R diagram.
- It is a static diagram.
- Shown using class as its basic entity and lines between them represents relationship between them.
- Describes the attributes and operations of a class and also the constraints imposed on the system.

Class Diagram for Customer Order



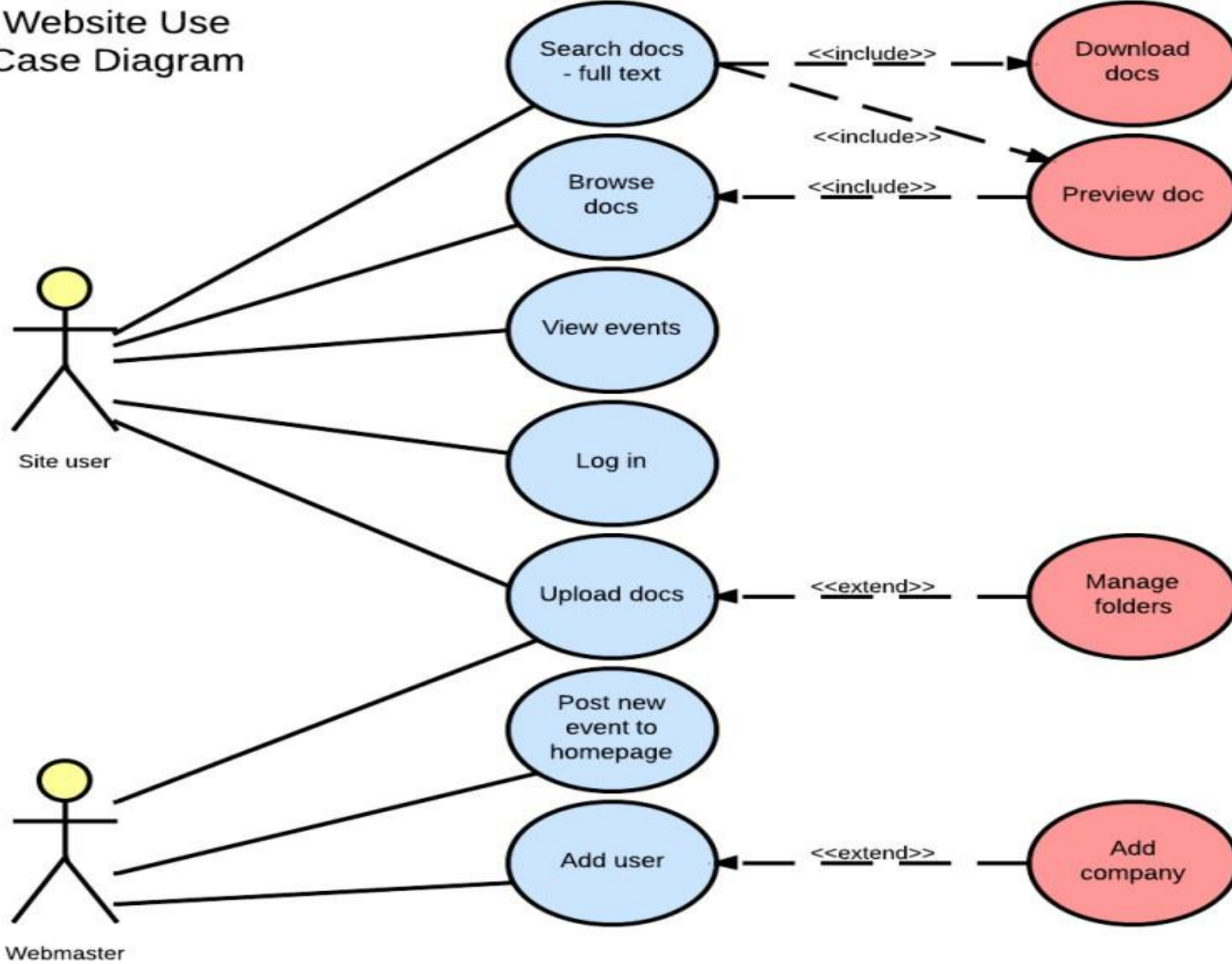
Class Diagram for Order Processing System



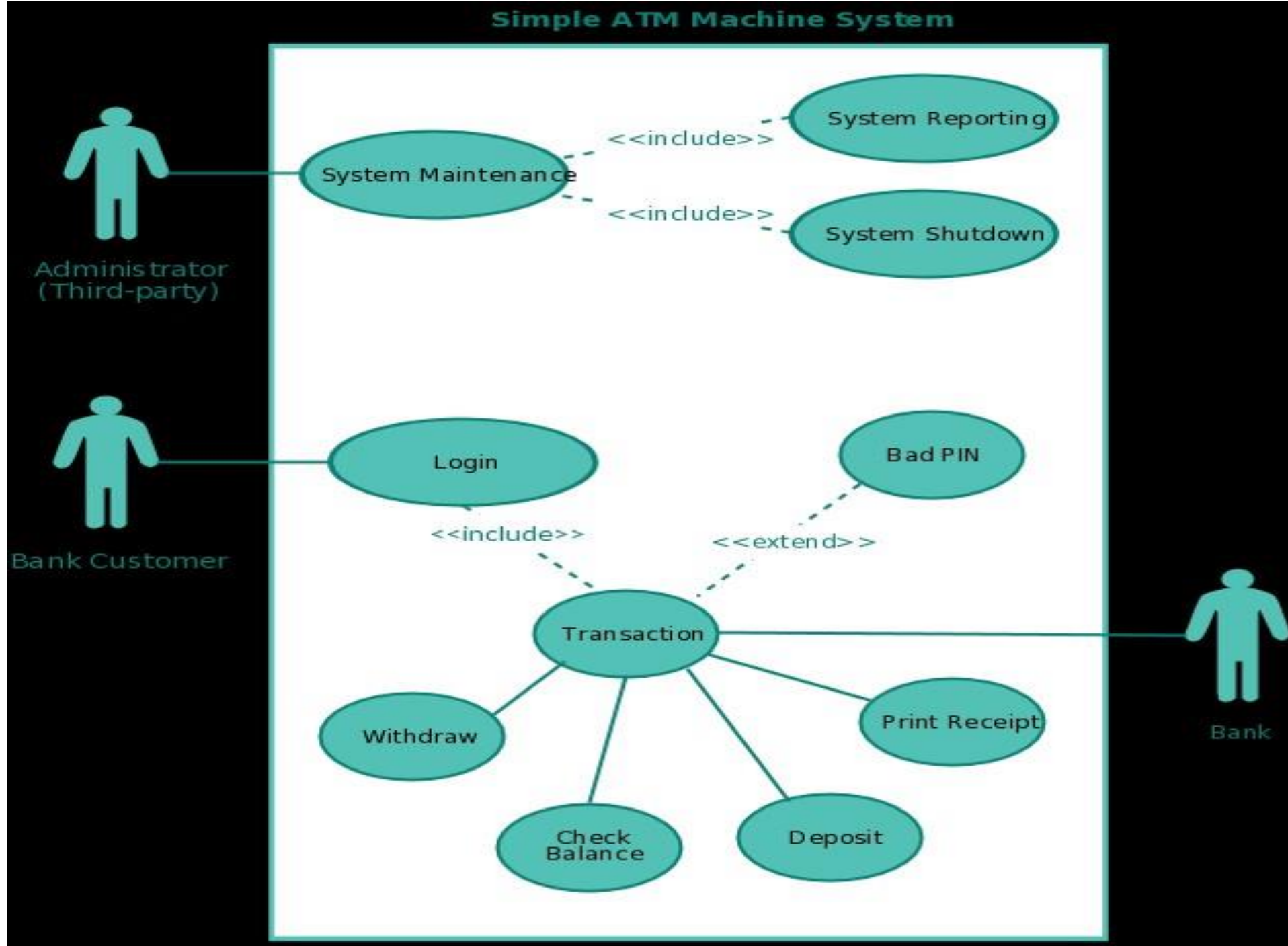
Use case diagram

- Shows the interaction between users and the system in particular steps of tasks that users perform.
- Purpose is to capture the dynamic aspect of a system.
- Purpose is to gather requirements of a system and to identify external and internal factors influencing the system.

Website Use Case Diagram



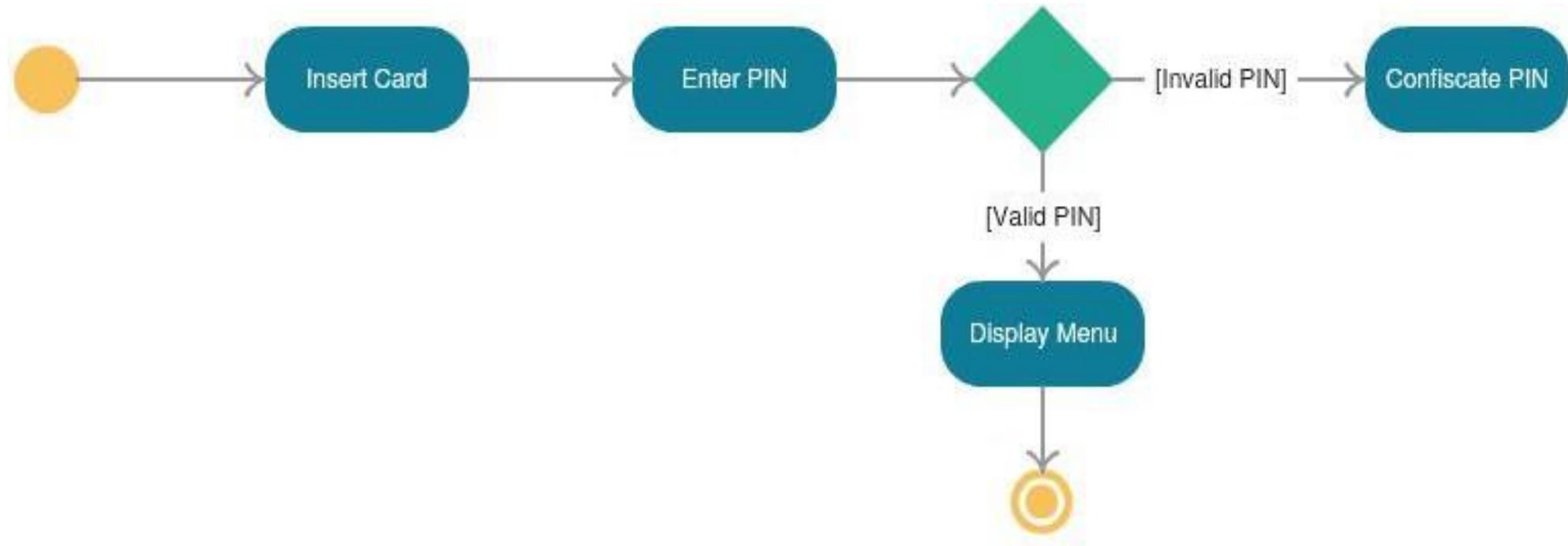
Use case
Diagram
for
User1-
Admin
User 2-
Customer



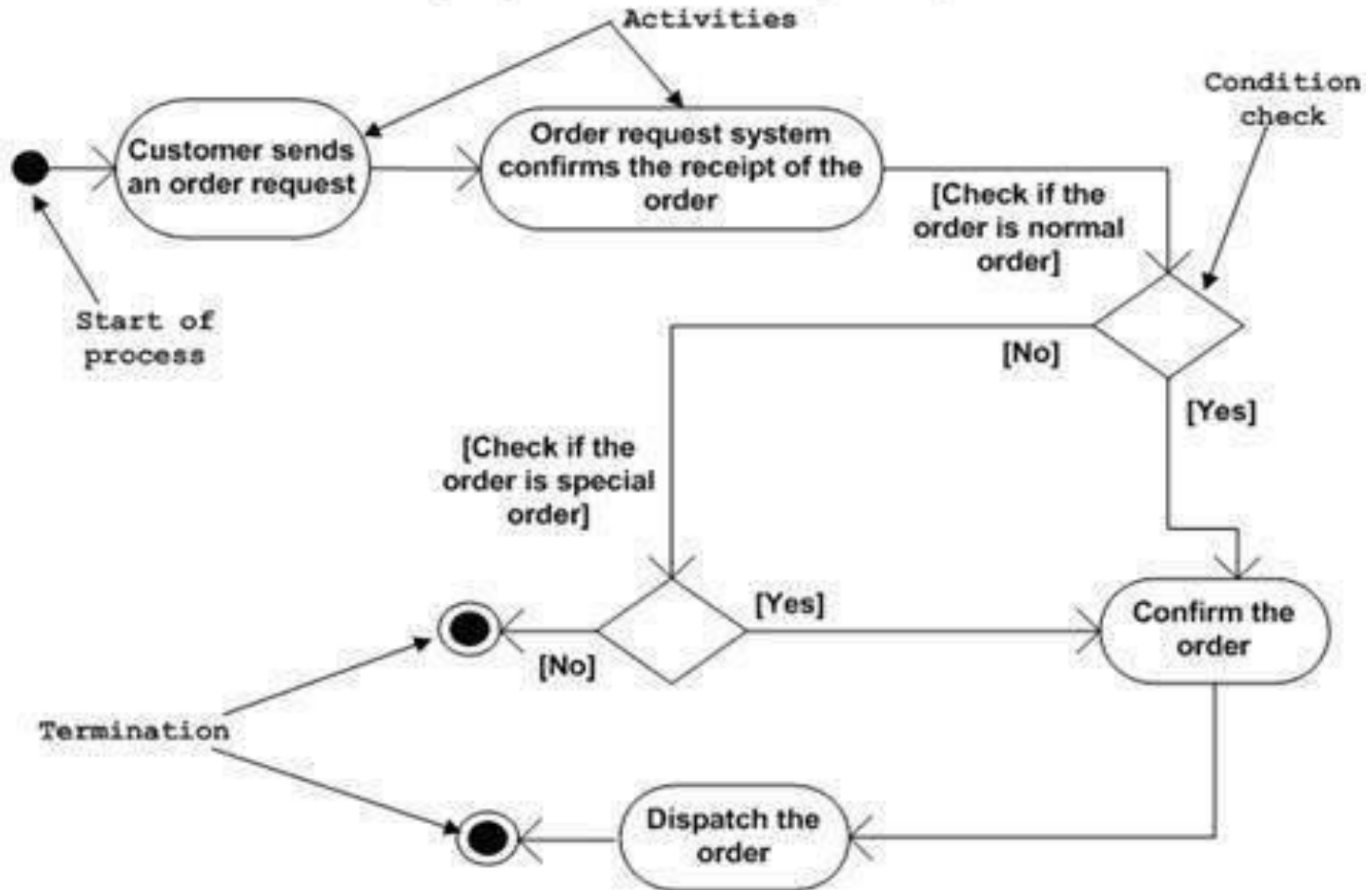
Activity Diagram

- Is basically a flowchart to represent the flow from one activity to another. The activity can be described as an operation of the system.
- It is a particular operation of the system.
- Are not only used for visualizing dynamic nature of a system but also used to construct the executable system by using forward and reverse engineering techniques.

Activity Diagram for ATM Card

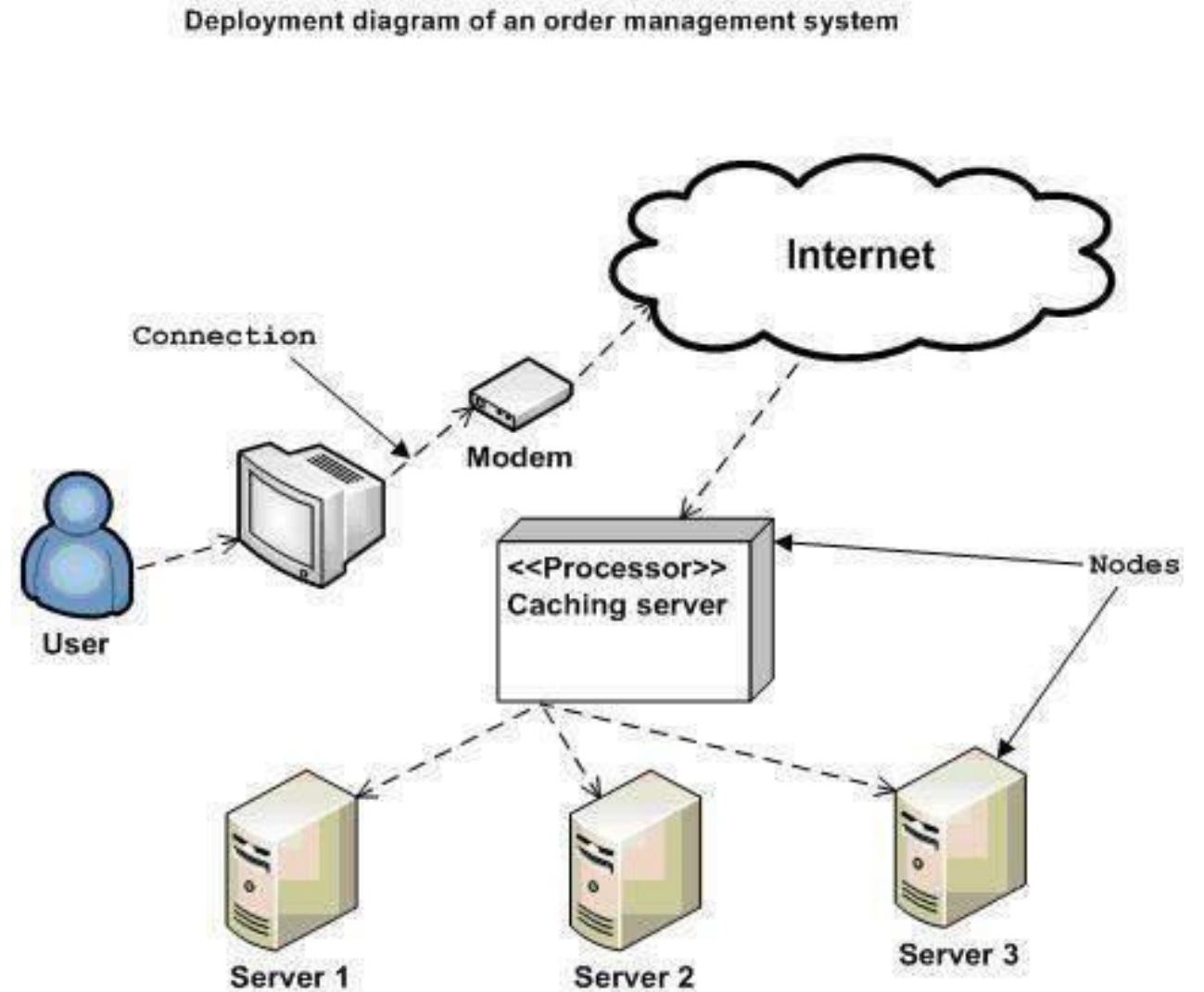


Activity diagram of an order management system



Implementation diagram

Are deployment diagrams which are used for describing about the hardware components where software components are deployed.



Advantages Of UML Diagrams

- It is the most useful method of visualization and documenting software systems design.
- It is effective for modeling large, complex software systems.
- It is simple to learn, but provides advanced features for expert analysts, engineers, designers and architects.
- It can specify systems in an implementation- independent manner.
- It specifies a skeleton that can be refined and extended with additional features.
- It specifies the functional requirements of system in an object oriented manner.

Disadvantages Of UML Diagrams

- Still no specification for modeling of graphical user interface.
- Poor for distributed systems – no way to formally specify serialization and object persistence.

Thank You!!!

